Electrochemical Wound Biosensor: Device Modelling Report

17-Layer End-to-End Computational Pipeline

Eton iGEM 2026, Dry Lab | MMP-9 / HNE Ratiometric SWV Sensor | C15 peptide substrate, Methylene blue reporter

What This Sensor Does

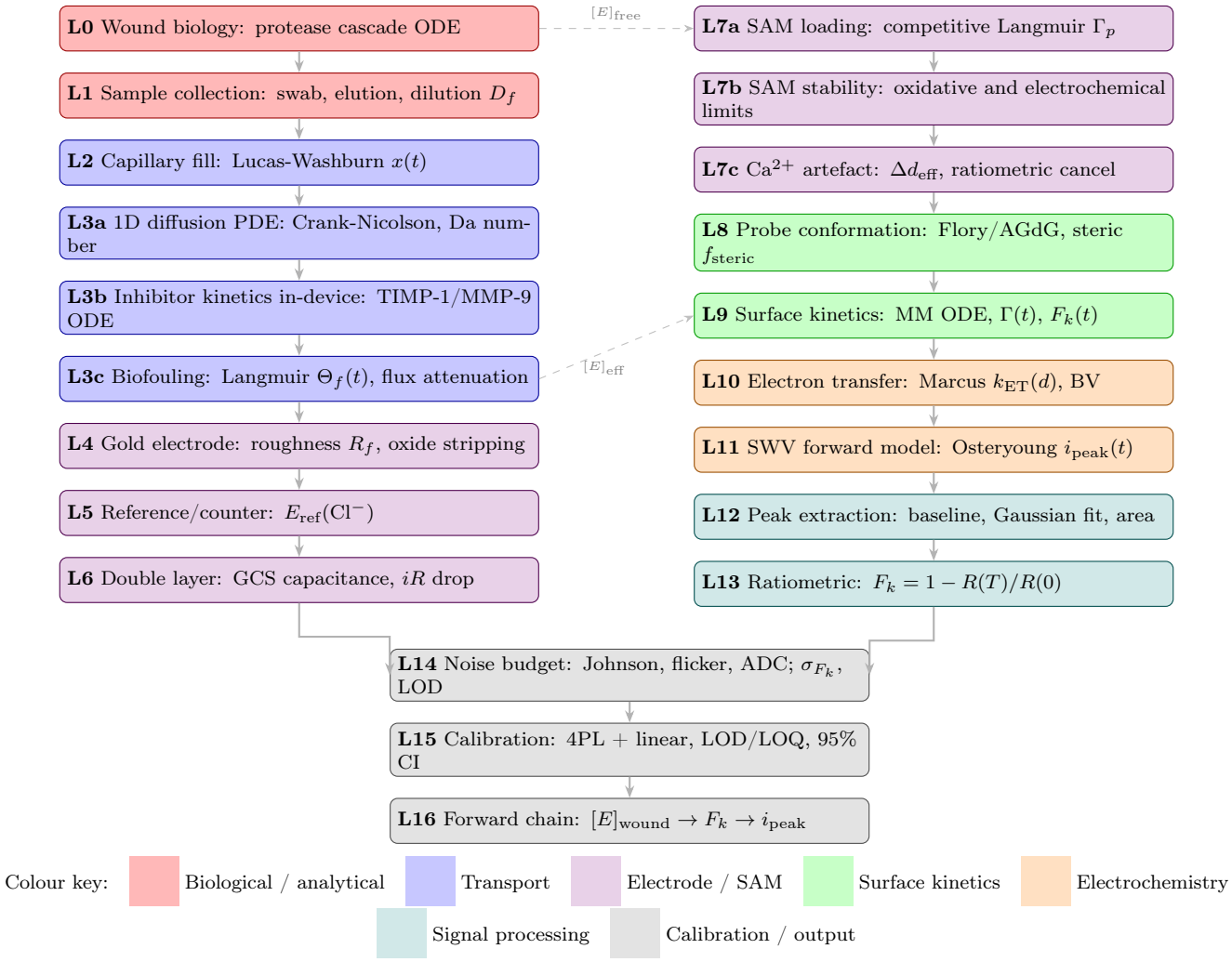
The sensor detects MMP-9 in fluid from a chronic wound. MMP-9 levels above 17 ng/mL indicate a wound that is failing to heal. The device works by coating a gold electrode with short peptide chains that MMP-9 can cut. The more MMP-9 is present, the more chains get cut, and the lower the electrical signal. A second electrode coated with an MMP-9-resistant chain acts as a reference: the ratio of the two signals cancels out background noise from temperature, salt, and fouling. The final output is F_k (cleavage fraction) – the proportion of probe chains cut during a fixed incubation window.

This report models every physical step from the wound to the electrical reading. Each section begins with a plain description of what that step calculates and why it matters, followed by the underlying equations. **Amber boxes** mark values that are currently estimated rather than directly measured; the full list of measurements needed, with suggested sources, is in Table 1 (Section 20). **Green boxes** summarise confirmed findings.

Variables and Symbols

Symbol	Plain-English meaning	Typical value	Unit
[MMP-9]	Concentration of MMP-9 enzyme in wound fluid	17–200 (clinical)	ng/mL
$[E]$	Same as [MMP-9], expressed in molar units	0.19–2.17 (clinical)	nM
F_k	Fraction of probe chains cut by MMP-9 (the sensor signal)	0 to 1	–
Γ	Number of probe chains attached per cm^2 of electrode	3 (design target)	pmol/cm^2
Γ_0	Initial probe density before any cleavage occurs	3	pmol/cm^2
k_{cat}/K_m	How fast MMP-9 cuts the C15 probe in free solution	188,000	$M^{-1}s^{-1}$
$k_{\text{eff,surf}}$	Surface second-order rate constant for MMP-9 cleaving the electrode-bound probe; must be measured (see Layer 9)	unknown	$M^{-1}s^{-1}$
$K_{m,\text{surf}}$	Probe density at which cleavage rate is half its maximum	3.4	pmol/cm^2
T	Incubation time (how long we wait before reading)	30–60	min
D_f	Dilution factor from eluting wound fluid off the swab	2	–
η	Fraction of each species successfully eluted from the swab	0.65–0.90	–
i_{peak}	Electrical current at the peak of the SWV scan	∼5	nA
R	Active/control electrode current ratio; cancels fouling and drift	0 to 1	–
σ_{F_k}	Measurement noise in the cleavage fraction	0.003–0.5	%
LOD	Lowest MMP-9 concentration the sensor can *detect* (3σ criterion)	8.8	ng/mL
LOQ	Lowest concentration that can be reliably *quantified* (10σ criterion)	29.4	ng/mL
TIMP-1	Natural MMP-9 inhibitor present in wound fluid; reduces free MMP-9	200 (wound)	ng/mL
A1AT	Natural HNE inhibitor; rapid and nearly complete HNE neutralisation	100 (wound)	μg/mL
f_{steric}	Fraction of probes that an MMP-9 molecule can physically reach	9–40%	–
k_{ET}	Rate at which methylene blue transfers electrons to the electrode	depends on distance	s^{-1}

Pipeline at a Glance



Key results (C15 substrate; F_k values are at the solution upper bound $k_{eff,surf} = 188{,}000$ $M^{-1}s^{-1}$ and scale proportionally with the true surface value, which must be measured).

- F_k at 17 ng/mL: **1.0%** at 30 min, **2.0%** at 60 min (upper bound). At 200 ng/mL: **11.4%** at 30 min, **21.9%** at 60 min (upper bound).
- **4PL calibration is required** – no valid linear window exists at any keff that also achieves detection at 17 ng/mL.
- Electronics noise floor: $\sigma_{F_k}^{elec} = 0.003\%$ – electronics are **not** the limiting factor.
- LOD (from calibration, using $\sigma_{F_k}^{cal} = 0.5\%$): **8.8 ng/mL** at $T = 30$ min at solution upper bound. Scales inversely with $k_{eff,surf}$.
- TIMP-1 strongly suppresses MMP-9 inside the device: >70% of MMP-9 captured within 30 min at wound-fluid TIMP-1 concentrations.
- C15 preferred over A11: 65% vs 35% F_k at 200 ng/mL, 60 min, solution upper bound.

Layer 0: How Much MMP-9 is in the Wound?

This step models the chemistry happening in the wound fluid itself. Two types of enzymes are present: MMP-9 (the target we detect) and HNE (used for the second sensor channel). Both are partially or fully blocked by natural inhibitors. This layer predicts how much of each enzyme is in its free, active form at the time of swabbing.

The clinical range of 17–200 ng/mL refers to *total* MMP-9 in wound fluid. But the sensor only responds to the *free* fraction that can reach and cleave the probe. If TIMP-1 sequesters most of the MMP-9, the sensor signal will be much lower than the total concentration implies.

MMP-9 activation and inhibition (9-state ODE):

$$\frac{d[\text{proMMP-9}]}{dt} = -\left(\frac{k_{\text{cat}}}{K_m}\right)_{\text{MMP-3}}[\text{MMP-3}][\text{proMMP-9}] - \left(\frac{k_{\text{cat}}}{K_m}\right)_{\text{plasmin}}[\text{plasmin}][\text{proMMP-9}]$$

$$\frac{d[\text{MMP-9}]_{\text{free}}}{dt} = \text{activation} - k_{\text{on,T1}}[\text{MMP-9}][\text{TIMP-1}] + k_{\text{off,T1}}[\text{complex}]$$

$$\frac{d[\text{HNE}]_{\text{free}}}{dt} = -k_{\text{on,A1AT}}[\text{HNE}][\text{A1AT}] - k_{\text{on,SLPI}}[\text{HNE}][\text{SLPI}]$$

Confident parameters: $k_{\text{on,A1AT}} = 2 \times 10^7$ $\text{M}^{-1}\text{s}^{-1}$ (Beatty 1980; Bieth 1984); K_i(TIMP-1/MMP-9) = 0.1 nM (Nagase & Woessner 1999).

What we still need to measure:

- (k_{cat}/K_m) for MMP-3 activating pro-MMP-9 (currently 5,000 $\text{M}^{-1}\text{s}^{-1}$, estimated) – search MEROPS database and Suzuki 1990 *J Biochem.*
- (k_{cat}/K_m) for plasmin activating pro-MMP-9 (currently 10,000 $\text{M}^{-1}\text{s}^{-1}$, estimated) – search fibrinolysis/MMP activation literature.
- $[\text{TIMP-1}]_{\text{wound}}$ = 200 ng/mL from Trengove 1999 – confirm whether this is free TIMP-1 or total (including TIMP-1 already bound to MMP-9). Free fraction matters here.
- $[\text{A1AT}]_{\text{wound}}$ = 100 μg/mL – this is estimated from serum dilution. Need a wound-fluid specific measurement. Proteomics or ELISA on chronic wound exudate.
- $k_{\text{on,SLPI}}$ for SLPI binding HNE (currently 10^5 $\text{M}^{-1}\text{s}^{-1}$, estimated) – SLPI is a slow tight-binding inhibitor; check serpinDB.
- $k_{\text{on,TIMP-1}}$ for TIMP-1 binding MMP-9 (currently 2×10^5 $\text{M}^{-1}\text{s}^{-1}$, Langton 1998) – confirm this is TIMP-1, not TIMP-2.
- See Table 1 (Section 20) for the full list with priority rankings.

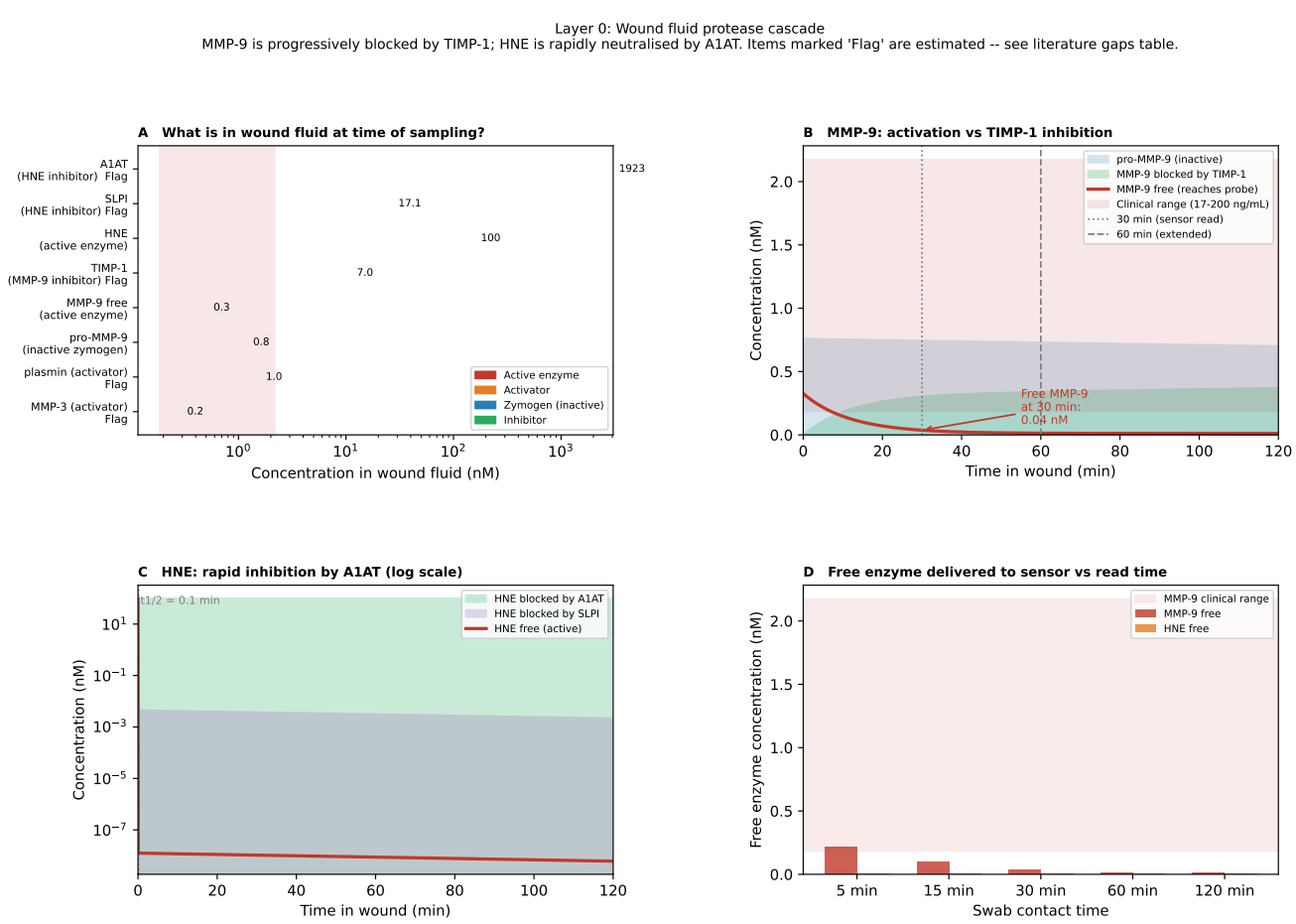


Figure 1: Layer 0: Wound-fluid protease cascade. **A:** Initial concentrations of all species (log scale). Inhibitors (green) greatly outnumber active enzymes (red). MMP-9 clinical detection range is shaded. **B:** MMP-9 free vs blocked over time. Free MMP-9 falls to ∼3.4 ng/mL by 30 min due to TIMP-1 capture – well below the 17 ng/mL total that enters the wound. **C:** HNE is completely neutralised by A1AT within seconds (log scale). **D:** Free enzyme concentration at different swab contact times: the earlier the swab is applied, the more free enzyme is captured.

Critical finding: the HNE channel will not function in real wound fluid.
At a wound-fluid A1AT concentration of 100 μg/mL (1.92 μM), the pseudo-first-order inhibition half-life of HNE is:

$$t_{1/2} = \frac{\ln 2}{k_{\text{on,A1AT}} \cdot [\text{A1AT}]} = \frac{0.693}{2 \times 10^7 \times 1.92 \times 10^{-6}} \approx 0.018 \text{ s}$$

HNE is irreversibly neutralised within approximately 0.02 seconds of entering the wound fluid. The model predicts free HNE at all time points beyond $t = 1$ s is effectively zero. No HNE reaches the sensor in measurable quantities.

Consequence for the sensor design. The device was designed as a ratiometric MMP-9/HNE sensor, with HNE providing the second channel. This ratiometric architecture is not viable in real wound fluid because the HNE signal will be zero regardless of wound state. Three alternative approaches should be evaluated before fabrication begins:

1. Replace the HNE active channel with a scrambled, protease-resistant peptide that serves as a purely electrochemical reference (cancels fouling and drift with no enzymatic activity).
2. Use a second MMP-9 substrate with a different cleavage rate to enable internal ratiometric calibration of MMP-9 activity alone.
3. Measure free HNE directly using a competitive immunoassay channel if HNE information is clinically required, rather than a cleavage-based electrode.

This conclusion is conditional on $[\text{A1AT}]_{\text{wound}} = 100$ μg/mL being correct. If the true chronic-wound A1AT concentration is substantially lower (<1 μg/mL, as might occur in a severely proteolytic environment), the HNE channel could recover. Measuring [A1AT] in patient-derived chronic wound exudate is therefore **CRITICAL priority** before committing to the HNE architecture.

Layer 1: Getting the Sample into the Device

The wound is swabbed to collect 30 μL of fluid, then the swab is placed in 60 μL of buffer to release the proteins (elution). This means the sample is diluted 2-fold before entering the sensor. Each protein elutes with a different efficiency η.

The dilution factor and elution efficiency together determine the actual concentration each species reaches inside the sensor. The more inhibitor (TIMP-1) co-elutes relative to MMP-9, the more the signal is suppressed inside the device.

$$[E]_{\text{chamber}} = [E]_{\text{wound}} \times \eta_E / D_f, \qquad D_f = 2 \tag{1}$$

What we still need to measure:
All species-specific elution efficiencies η listed below are estimated or borrowed from protein recovery studies in different matrices. Each needs a spike-and-recovery experiment on the actual swab/buffer combination:

- $\eta_{\text{MMP-9}} = 0.70$ (estimated); $\eta_{\text{TIMP-1}} = 0.75$; $\eta_{\text{A1AT}} = 0.80$; $\eta_{\text{SLPI}} = 0.85$ (Ashcroft 2000); $\eta_{\text{HNE}} = 0.70$ (estimated).

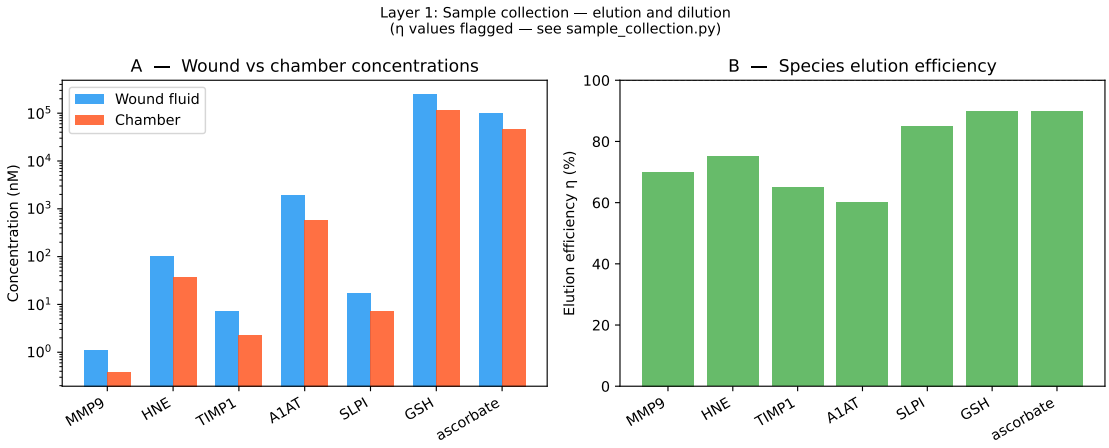


Figure 2: Layer 1: Wound vs chamber concentrations and per-species elution efficiency. MMP-9 undergoes 2.86× total dilution ($D_f = 2$, $\eta = 0.70$). All η values need experimental confirmation.

Layer 2: Filling the Sensor Channel

When the eluted sample is placed at the sensor inlet, it wicks into the channel by capillary action (the same force that makes paper absorb water). This step checks that the channel fills completely before measurement begins, and that the time to fill is much shorter than the incubation time.

If the channel does not fill completely, some electrodes will not be exposed to the sample. If filling takes longer than a minute, it eats into the usable incubation window.

$$x(t) = \sqrt{\frac{r_h\,\gamma\cos\theta}{2\eta_{\text{fluid}}}\cdot t}, \quad r_h = \frac{2wh}{w+h} \tag{2}$$

The Damkohler number Da $\approx 556 \gg 1$ confirms that enzyme transport to the electrode is not the slow step – kinetics (how fast MMP-9 cuts) dominate.

What we still need to measure:
Contact angle $\theta = 30°$ on PDMS/gold surface – measure on the actual chip.

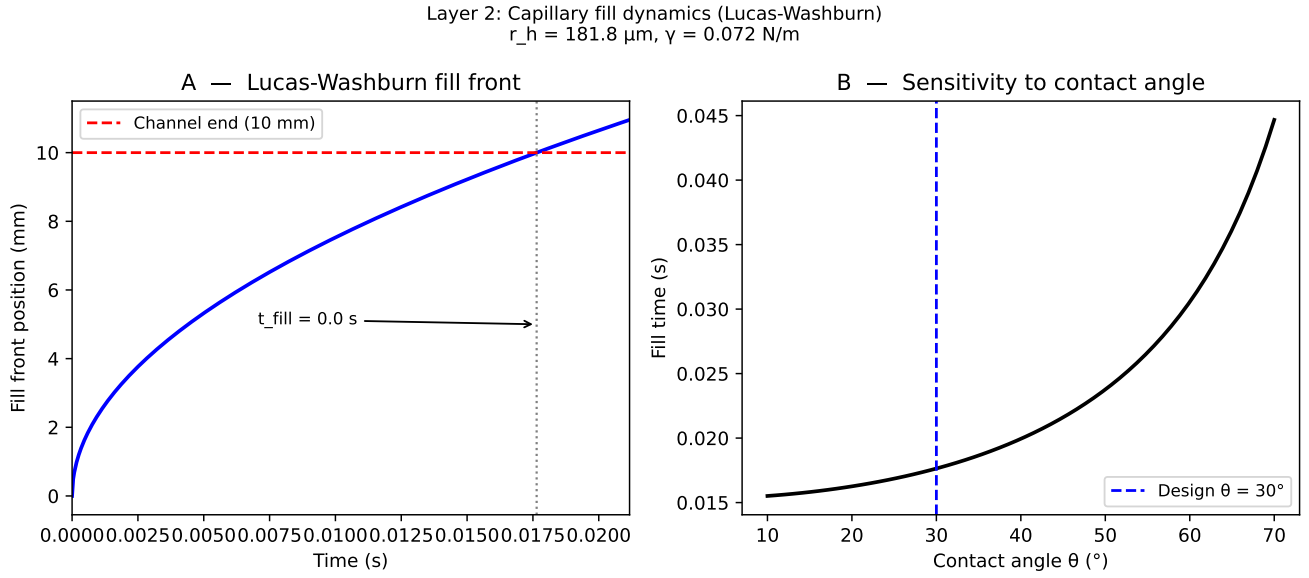


Figure 3: Layer 2: Capillary fill dynamics. The channel fills in <0.2 s – negligible compared to 30–60 min incubation. Da $\gg 1$ confirms kinetics are rate-limiting, not mass transport.

Layers 3a–3c: How the Sample Changes Inside the Device

3a: Enzyme diffusion to the electrode

Once inside the channel, MMP-9 diffuses toward the electrode surface where the probes are. This step checks that MMP-9 reaches the surface quickly relative to the incubation time, and that MMP-9 depletion at the surface is negligible (the concentration stays approximately uniform).

$$\frac{\partial [E]}{\partial t} = D_E \frac{\partial^2 [E]}{\partial z^2}, \quad D_{\text{MMP-9}} = 4.7 \times 10^{-11}\ \text{m}^2\text{s}^{-1} \tag{3}$$

MMP-9 diffusivity is calculated from the Stokes-Einstein equation using a radius of 3.5 nm.

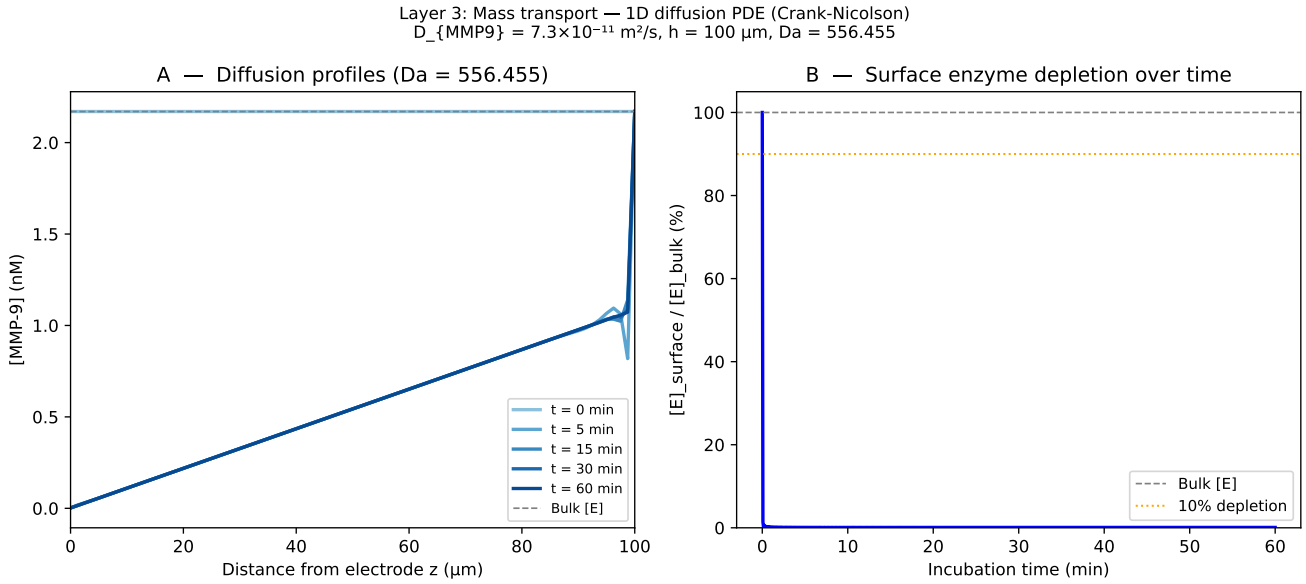


Figure 4: Layer 3a: MMP-9 diffusion in the sensor channel. The enzyme concentration remains nearly uniform throughout (surface depletion <1%) confirming diffusion is not the rate-limiting step.

3b: Inhibitor binding inside the device

After elution, TIMP-1 and MMP-9 are both in solution inside the device. TIMP-1 continues to bind MMP-9 during the incubation period, progressively reducing the free MMP-9 available to cleave probes. This is the *single largest source of signal suppression* in the model.

At 17 ng/mL MMP-9 (0.19 nM) and 200 ng/mL TIMP-1 (7 nM in wound fluid, ∼2.6 nM in chamber after dilution), TIMP-1 is in ∼13-fold molar excess. Over 70% of MMP-9 is captured within 30 min.

What we still need to measure:
$k_{\text{on,TIMP-1}} = 2 \times 10^5\ \text{M}^{-1}\text{s}^{-1}$ – confirm this applies to TIMP-1 (not TIMP-2). Confirm whether Trengove 1999 reports free or total TIMP-1 in wound fluid. If total, the model overestimates in-device inhibition.

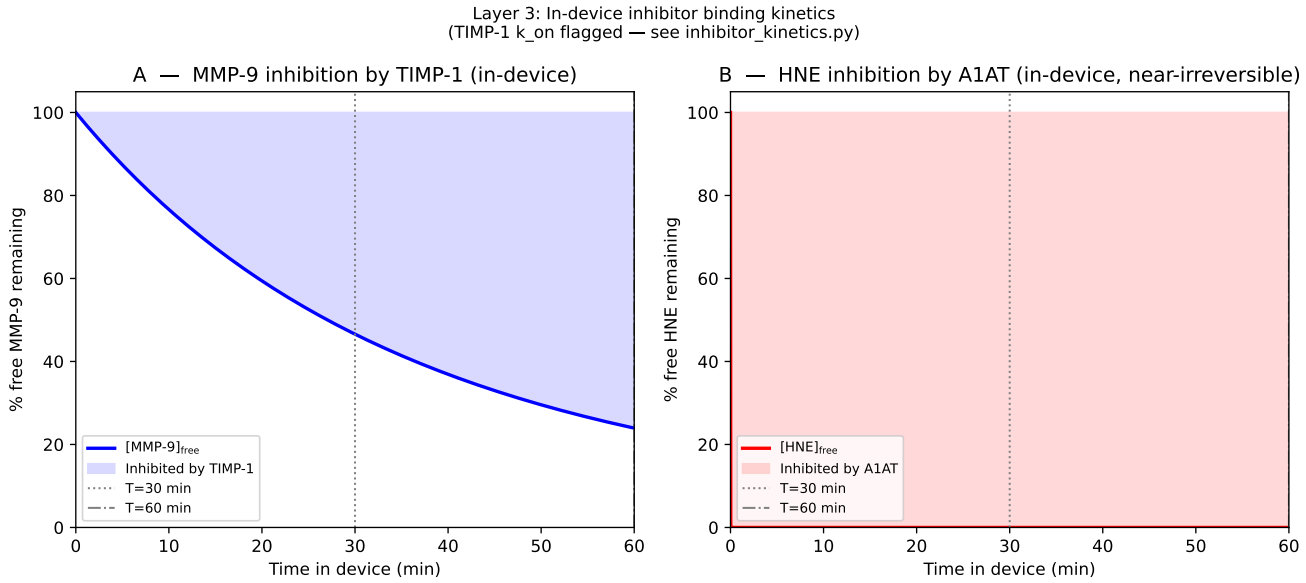


Figure 5: Layer 3b: In-device inhibitor binding. At clinical MMP-9 concentrations, the large TIMP-1 excess (∼13-fold) captures most MMP-9 within 30 min. HNE is quenched by A1AT within seconds. This inhibitor suppression is the dominant factor limiting the signal.

3c: Surface fouling

Wound fluid contains many non-specific proteins that can stick to the electrode surface, forming a physical barrier. This layer models how quickly that barrier builds up and how much it slows enzyme access to the probes.

$$\frac{d\Theta_f}{dt} = k_f\,[P]_{\text{bulk}}(1-\Theta_f) - k_d\,\Theta_f, \qquad \alpha_{\text{foul}} = \frac{h_{\text{channel}}}{h_{\text{channel}} + h_{f,\max}\Theta_f} \tag{4}$$

The model predicts fouling has <0.01% effect on F_k because the fouling layer ($h_f \sim 10$ nm) is negligible relative to the 100 μm channel height. The ratiometric design also cancels fouling if both active and control electrodes foul equally.

What we still need to measure:
$k_f = 10^3$ M^{-1}s^{-1}, $k_d = 10^{-4}$ s^{-1} – both estimated. Measure by QCM in wound-fluid matrix. The protein diffusivity through a dense fouling layer is also assumed equal to bulk (could be 10–100× lower – see audit finding 29).

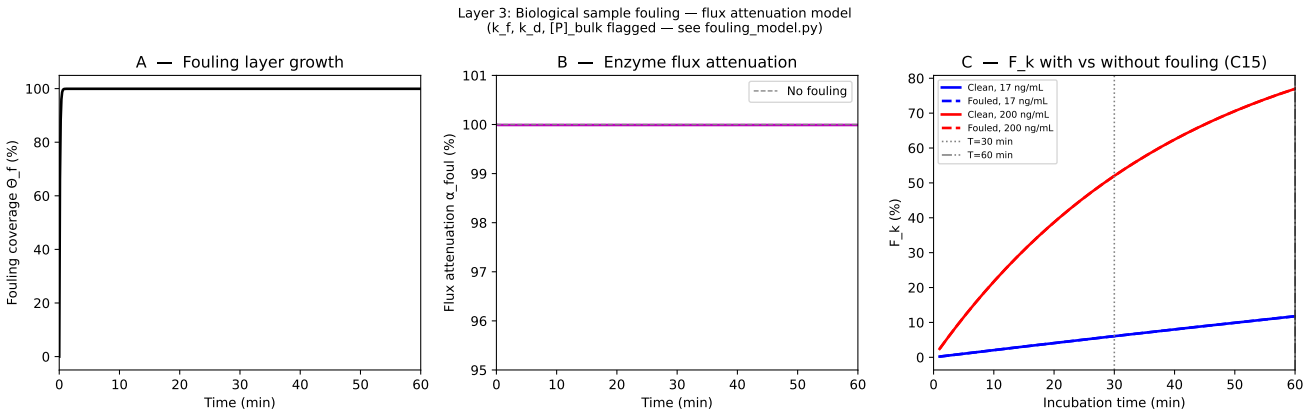


Figure 6: Layer 3c: Biofouling coverage $\Theta_f(t)$ and flux attenuation. Fouling saturates quickly but its geometric effect on signal is <0.01%.

Layers 4–7: The Electrode Surface and SAM Chemistry

Before any experiment, the gold electrode is coated with a self-assembled monolayer (SAM): a carpet of short molecules that hold the peptide probes upright and provide a clean electrical surface. Layers 4–7 model the preparation and stability of this surface.

Layer 4: Gold electrode

True electrode area (accounting for surface roughness) is measured from the gold oxide stripping charge: $A_{\text{true}} = R_f \times A_{\text{geo}}$, where $Q_{\text{ref}} = 390\ \mu\text{C}\,\text{cm}^{-2}$ (Trasatti & Petrii 1992).

What we still need to measure:
Roughness factor $R_f = 1.5$ (nominal). Measure by recording the oxide stripping charge on the actual electrode. Note: R_f is not yet propagated into the steric or kinetic models (see audit finding 15).

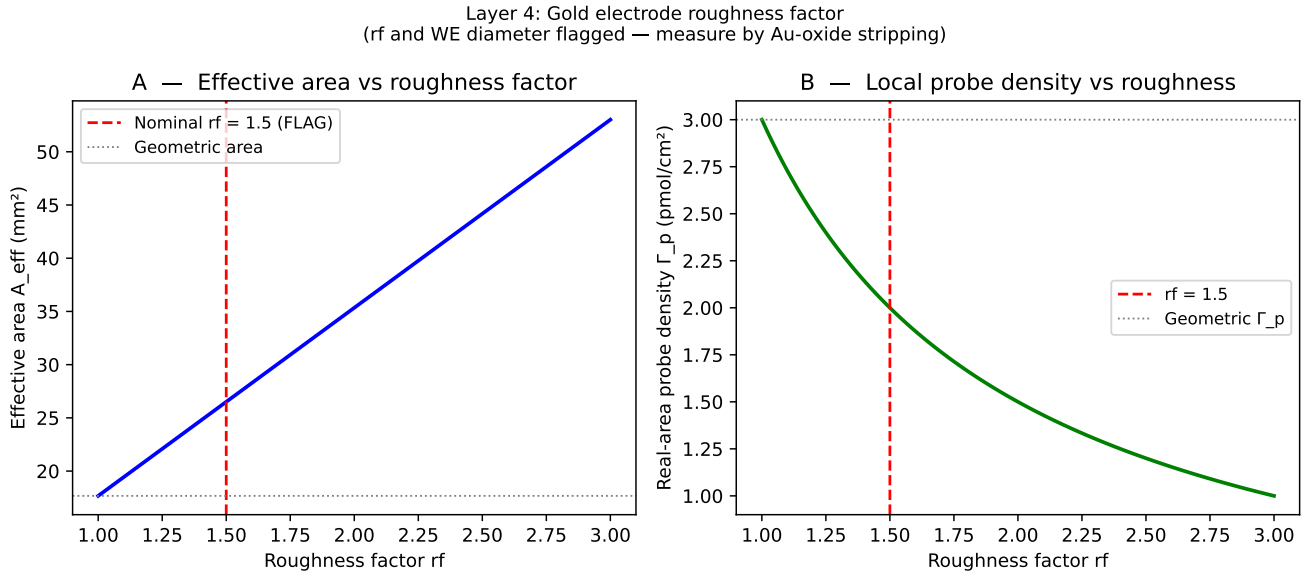


Figure 7: Layer 4: Roughness factor calculation and effective area vs oxide stripping charge.

Layer 5: Reference electrode

On-chip Ag/AgCl reference: $E^{\circ}_{\text{AgCl}} = 0.2224$ V vs SHE; $[\text{Cl}^-]_{\text{wound}} = 150$ mM. Reference drift from wound-to-chamber Cl^- change: $\Delta E_{\text{ref}} = 1.5$ mV – negligible.

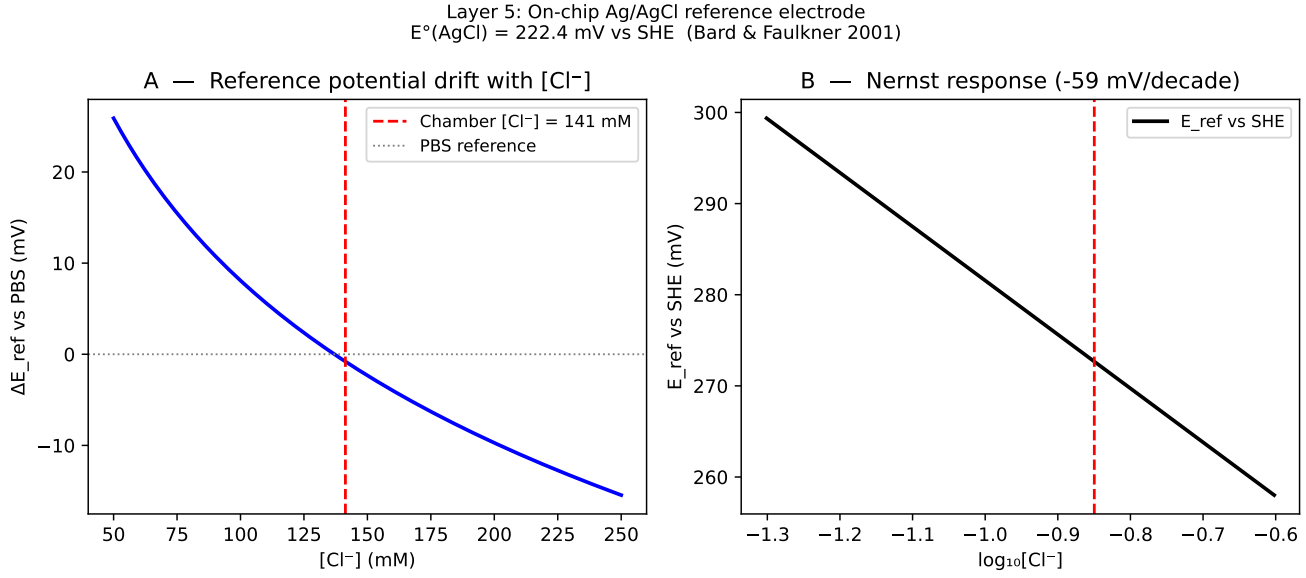


Figure 8: Layer 5: Reference electrode potential vs Cl^-. Shift from wound to chamber is <2 mV.

Layer 6: Double layer capacitance

The double layer (the region of charge at the electrode-solution interface) acts like a capacitor in parallel with the faradaic (electron-transfer) signal. At 163 mM ionic strength, double-layer charging current in SWV is ∼0.6 nA – well below the ∼5 nA MB peak.

What we still need to measure:
Helmholtz capacitance $C_H = 30\ \mu\mathrm{F\,cm^{-2}}$ assumed for MCH-SAM. Measure by electrochemical impedance spectroscopy (EIS) on the actual surface.

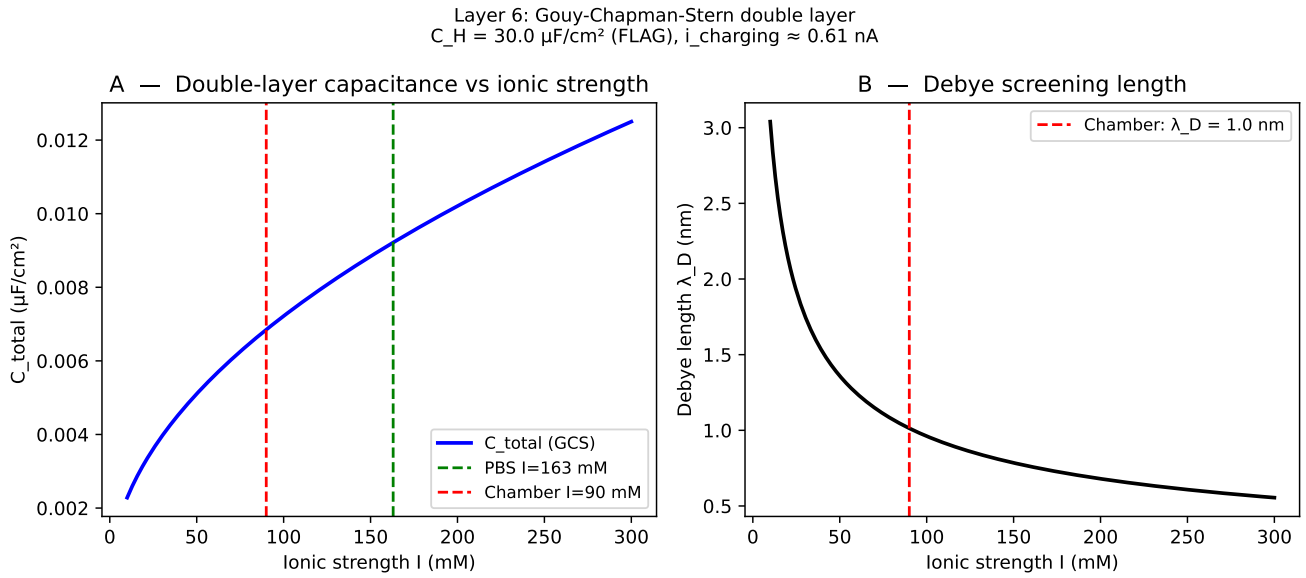


Figure 9: Layer 6: Double-layer model. Charging current is well below the faradaic signal across all ionic strengths encountered.

Layer 7: SAM formation and stability

The SAM is deposited by soaking the electrode in a mixture of the probe molecule and a backfiller (MCH). Probe density Γ_p is controlled by adjusting the ratio of probe to MCH in the deposition solution:

$$\frac{\Gamma_p}{\Gamma_{\max}} = \frac{K_{a,p}\, c_p}{1 + K_{a,p}\, c_p + K_{a,d}\, c_d} \tag{5}$$

Target: $\Gamma_p = 2$–$3\ \mathrm{pmol\,cm^{-2}}$. Above ∼3.3 $\mathrm{pmol\,cm^{-2}}$, MMP-9 molecules are too tightly packed to fit between probes (see Layer 8).

SAM electrochemical stability: scan minimum −0.65 V vs desorption threshold −0.90 V – 250 mV margin. Oxidative loss at 10 μM H_2O_2 over 60 min: 3.5% – acceptable.

What we still need to measure:
Adsorption constants $K_{a,p}$, $K_{a,d}$ – must be measured by SPR or QCM during SAM formation to predict the solution composition needed for target Γ_p.

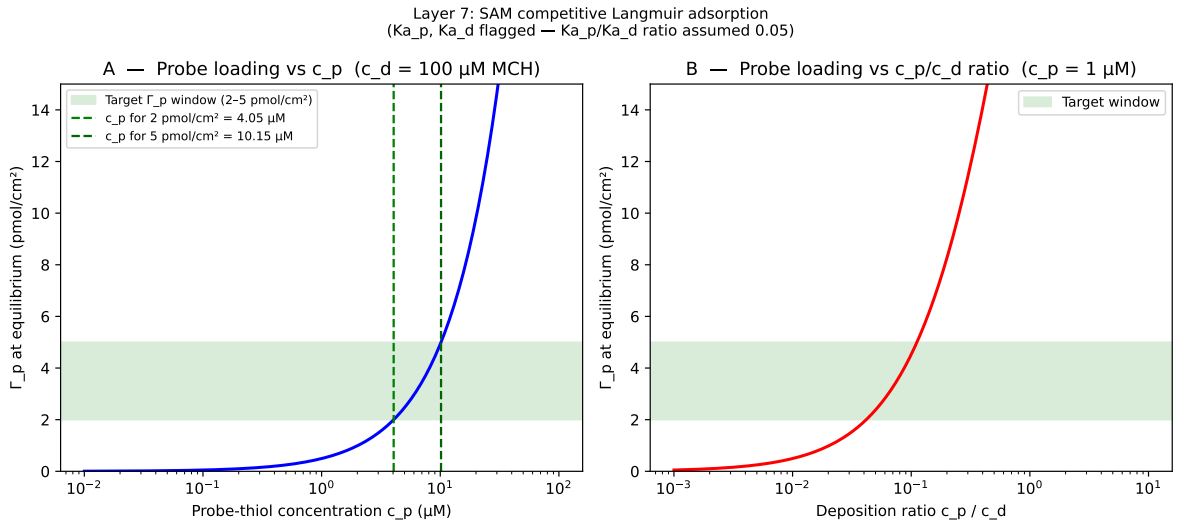


Figure 10: Layer 7a: SAM loading. Probe density vs solution composition. At $c_p \approx 6$ μM and $c_d = 100$ μM MCH the model predicts $\Gamma_p \approx 3$ pmol cm^{-2} (all values flagged – $K_{a,p}$, $K_{a,d}$ must be measured).

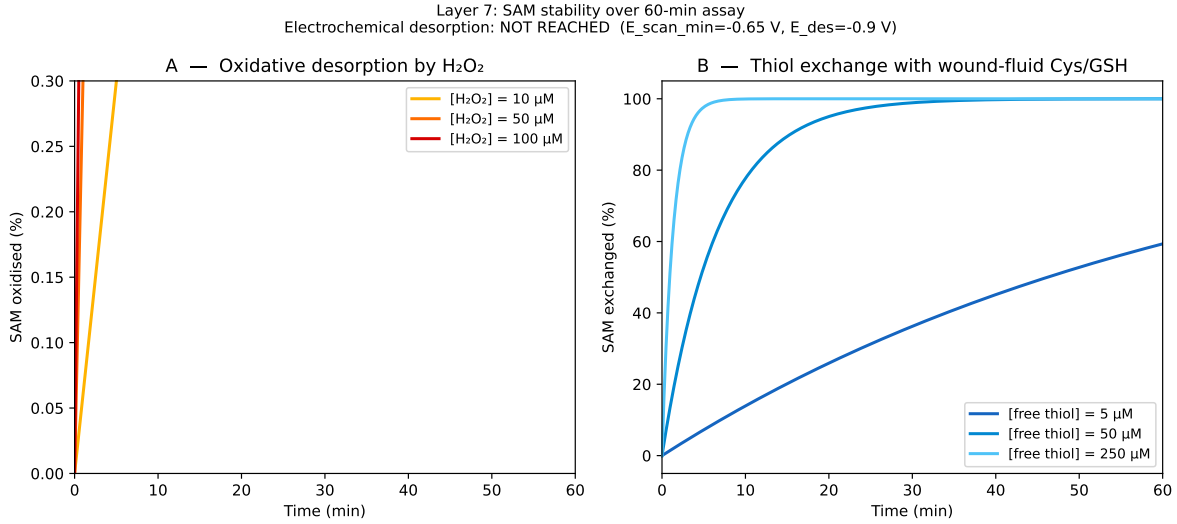


Figure 11: Layer 7b: SAM stability under oxidative and electrochemical stress. Desorption potential margin: 250 mV. Oxidative loss at 10 μM H_2O_2, 60 min: 3.5%.

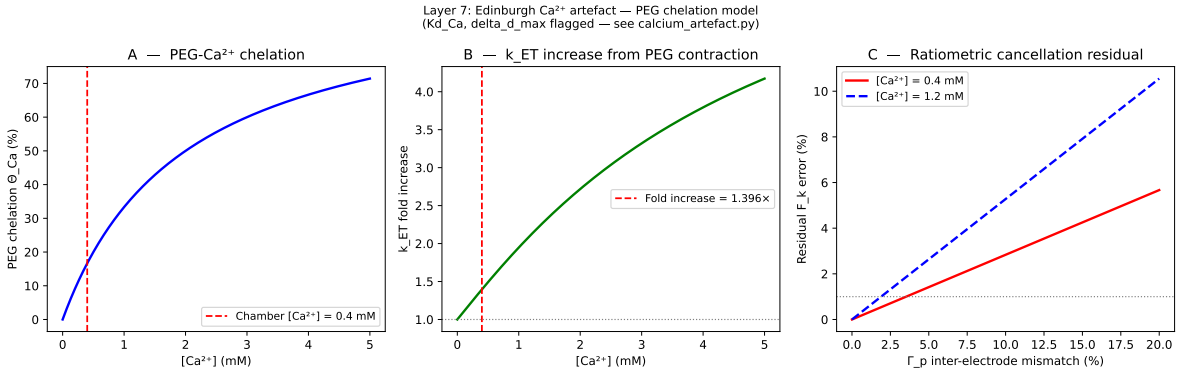


Figure 12: Layer 7c: Ca^{2+} artefact. PEG chelates Ca^{2+} ($K_d \approx 2$ mM), contracting the tether and increasing k_{ET}. The ratiometric design cancels this artefact *provided* both active and control electrodes have identical Γ_p.

Layer 8: Can MMP-9 Actually Reach the Probe?

MMP-9 is a large enzyme (4 nm radius). If the probes are packed too densely, MMP-9 cannot physically squeeze between them to cleave the probe it is trying to reach. This layer calculates the fraction of probes that are accessible.

At the target density of 3 pmol cm^{-2}, only 9% of probes are accessible to MMP-9. This dramatically reduces the signal. If density is lowered to 1 pmol cm^{-2}, accessibility rises to ∼70%.

$$f_{\text{steric}} = 1 - \frac{\Gamma_p}{\Gamma_{\text{max}}^{\text{steric}}}, \qquad \Gamma_{\text{max}}^{\text{steric}} = \frac{1}{\pi r_E^2 N_A} \approx 3.3 \text{ pmol cm}^{-2} \tag{6}$$

At $\Gamma_p = 3$ pmol cm^{-2}: $f_{\text{steric}} = 9\%$. At $\Gamma_p = 1.5$ pmol cm^{-2}: $f_{\text{steric}} = 54\%$. **Design recommendation: aim for $\Gamma_p \leq 2$ pmol cm^{-2}.**

This is one of the most important design parameters. Packing the probes too densely causes the enzyme to bounce off without cleaving. The model suggests the optimal probe density is around 1.5–2 pmol/cm^2, not 3 pmol/cm^2.

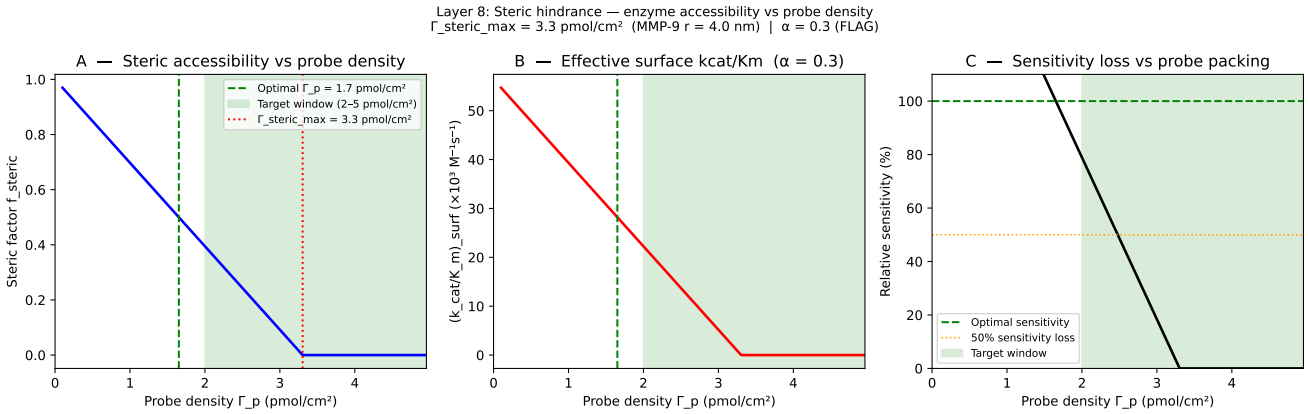


Figure 13: Layer 8: Steric accessibility. At the target density of 3 pmol cm^{-2}, only 9% of probes are reachable by MMP-9. Reducing to 1.5–2 pmol cm^{-2} improves accessibility to 50–70% while maintaining adequate signal amplitude.

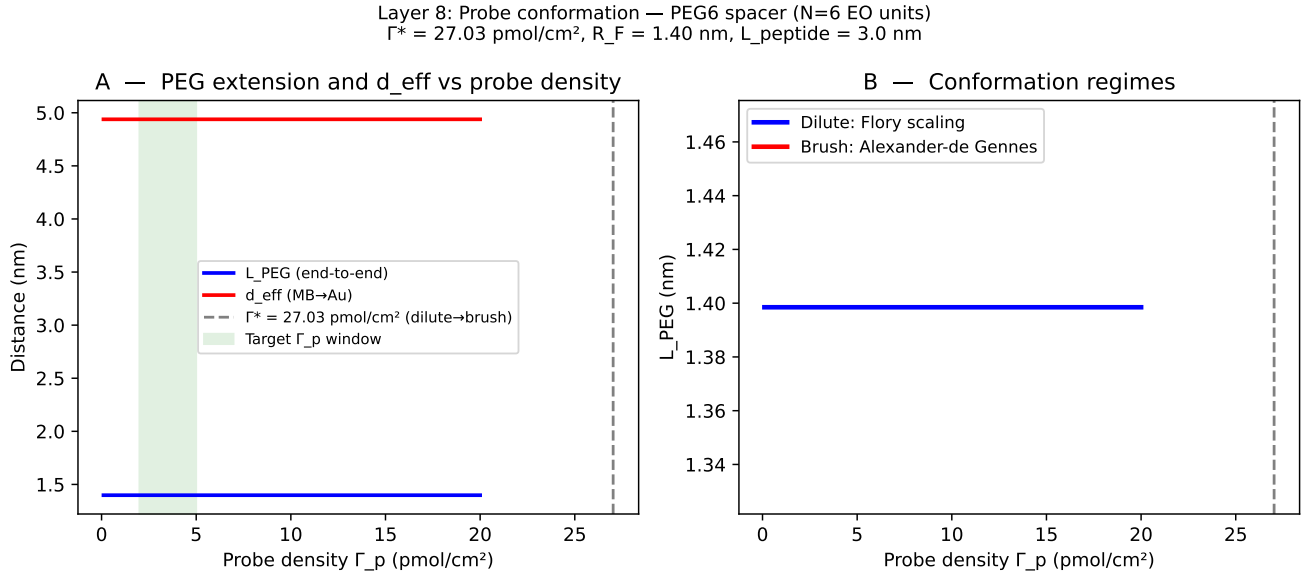


Figure 14: Layer 8: Probe conformation model. PEG$_6$ effective length $L_{\text{PEG}} = 1.4$ nm in the dilute Flory regime. Total probe-to-electrode distance $d_{\text{eff}} \approx 4.9$ nm.

Layer 9: Surface Enzyme Kinetics

This is the central calculation of the pipeline. It models how fast MMP-9 cuts the surface probe chains and how F_k evolves with enzyme concentration and incubation time. Every downstream number – LOD, calibration curve shape, operating window – depends on this layer.

Three-state surface kinetic model

The model follows the framework of Anne, Demaille and Moiroux (2001), which treats the electrode as a distinct phase with its own rate constants. The mechanism is:

$$E + \Gamma_S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \Gamma_{ES} \xrightarrow{k_{\text{cat}}} E + \Gamma_P$$

where Γ_S (free probe), Γ_{ES} (enzyme-probe complex) and Γ_P (cleaved probe) are all surface densities in $\text{mol}\,\text{cm}^{-2}$, conserved as $\Gamma_S + \Gamma_{ES} + \Gamma_P = \Gamma_0$.

ODEs (Anne, Demaille, Moiroux 2001, Eq. 1):

$$\frac{d\Gamma_S}{dt} = k_{-1}\,\Gamma_{ES} - k_1\,[E]\,\Gamma_S \tag{7}$$

$$\frac{d\Gamma_{ES}}{dt} = k_1\,[E]\,\Gamma_S - (k_{-1} + k_{\text{cat}})\,\Gamma_{ES} \tag{8}$$

$$\frac{d\Gamma_P}{dt} = k_{\text{cat}}\,\Gamma_{ES} \tag{9}$$

k_1, k_{-1}, k_{cat} are **surface** rate constants – not solution values.
Two limiting cases (determined by k_{cat}/k_{-1}, currently unknown):

- **Quasi-equilibrium** ($k_{-1} \gg k_{\text{cat}}$): steady-state approximation gives $k_{\text{eff}} = k_{\text{cat}}/K_m$ where $K_m = (k_{-1} + k_{\text{cat}})/k_1$.
- **Irreversible** ($k_{-1} \ll k_{\text{cat}}$, or $k_1 \gg k_{-1}$): $k_{\text{eff}} = k_1$ (ES formation is rate-limiting).

Convergence at clinical concentrations

At clinical MMP-9 concentrations (0.19–2.17 nM) and any physically reasonable k_1:

$$k_1\,[E]_{\max} < 0.22\ \text{s}^{-1} \quad \ll \quad k_{\text{cat,sol}} = 703\ \text{s}^{-1}$$

The complex Γ_{ES} stays below 1 ppm of Γ_0 (Panel A below). Both kinetic limits therefore give the same cleavage fraction:

$$F_k(t) = 1 - \exp(-k_{\text{eff,surf}}\,[E]\,t)$$

$k_{\text{eff,surf}}$ is the surface second-order rate constant ($\text{M}^{-1}\text{s}^{-1}$). The solution value $(k_{\text{cat}}/K_m)_{\text{sol}} = 188{,}000\ \text{M}^{-1}\text{s}^{-1}$ (Max Li's C15 assay; $k_{\text{cat}} = 703\ \text{s}^{-1}$, $K_m = 3.74$ mM) is an upper bound for $k_{\text{eff,surf}}$.

What we still need to measure:

$k_{\text{eff,surf}}$ must be measured on the fabricated electrode. It is the single critical unknown in the model – every LOD and F_k prediction is conditional on it.
Measurement protocol: (1) prepare electrode with target SAM; (2) incubate at known $[E]$ for time T; (3) integrate the MB CV peak before and after to get Γ_0 and $\Gamma(T)$; (4) extract $k_{\text{eff,surf}} = -\ln(1 - F_k)/([E]\,T)$.
Note on steric hindrance (Layer 8): geometric crowding at the surface reduces how often an MMP-9 molecule reaches a probe, which reduces k_1 relative to its solution value. This effect is automatically captured within the measured $k_{\text{eff,surf}}$ and does not require a separate correction factor.
$\Gamma_0 = 3$ **pmol cm^{-2}** is a design target. Confirm by CV integration.

Predictions and operating window

Because $k_{\text{eff,surf}}$ is unknown, the model is run across the plausible range (2,000 to 188,000 $\text{M}^{-1}\text{s}^{-1}$) to bracket all outcomes. Three conclusions hold across the entire range:

- $F_k(t)$ is always exponential, never linear in t. A linear calibration in $[E]$ is valid only while $k_{\text{eff,surf}}\,[E_{\text{hi}}]\,T \ll 1$.
- At the solution upper bound, the sensor detects 17 ng/mL in <20 min; the linear calibration window closes around $T = 16$ min.

- At a conservative 10,000 $\mathrm{M}^{-1}\mathrm{s}^{-1}$, detection at 17 ng/mL requires $T > 128$ min.
- **4PL calibration should be used regardless of keff**, as the linear approximation breaks at different times for different surface chemistries.

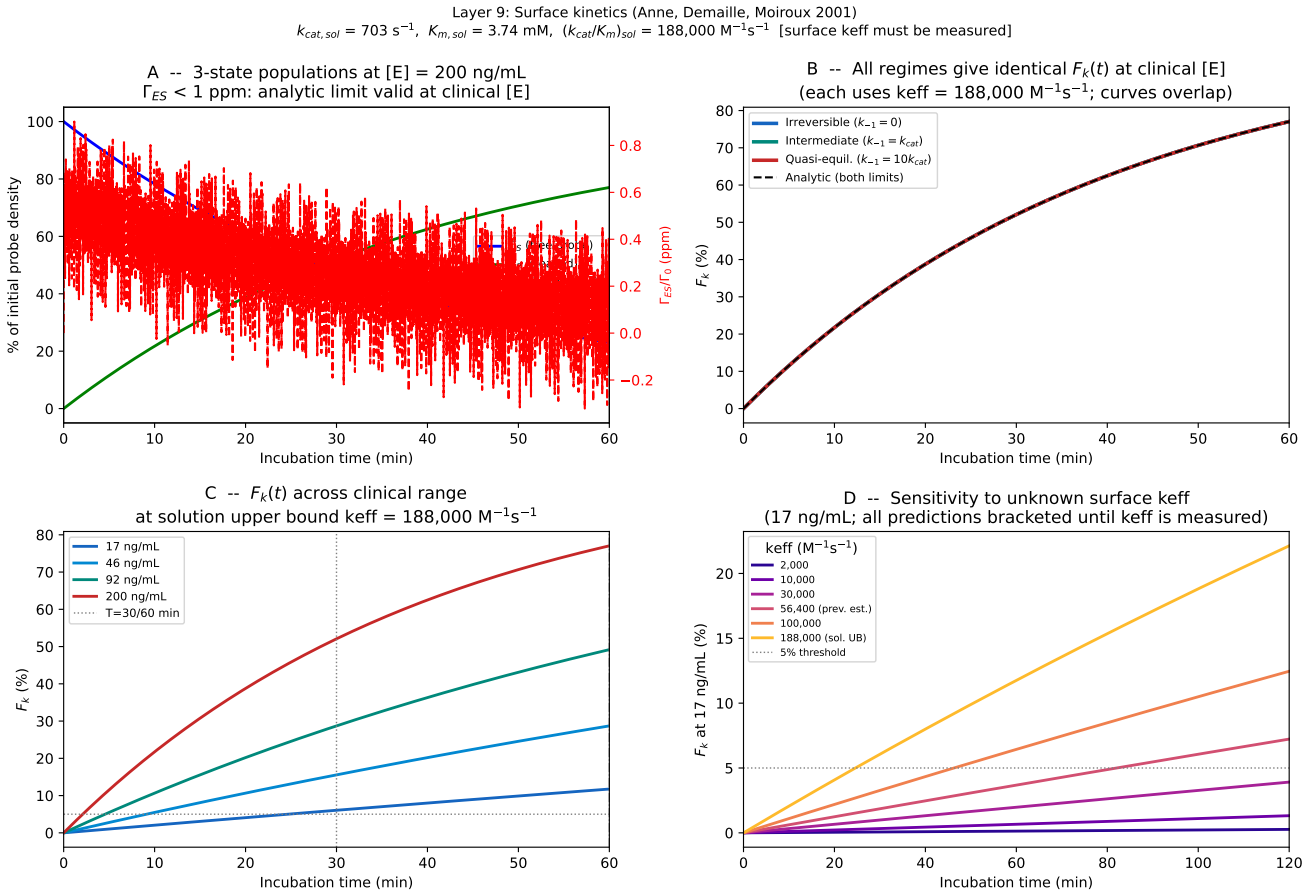


Figure 15: Layer 9: Three-state surface kinetic model (Anne, Demaille, Moiroux 2001). **A:** Γ_S, Γ_P and Γ_{ES} (right axis, ppm scale) at 200 ng/mL – the complex accumulates to less than 1 ppm, confirming the analytic limit. **B:** All three kinetic regimes give identical $F_k(t)$ at clinical [E]. **C:** $F_k(t)$ across the clinical range at the solution upper bound keff. **D:** Sensitivity to the unknown surface keff – predictions span two orders of magnitude.

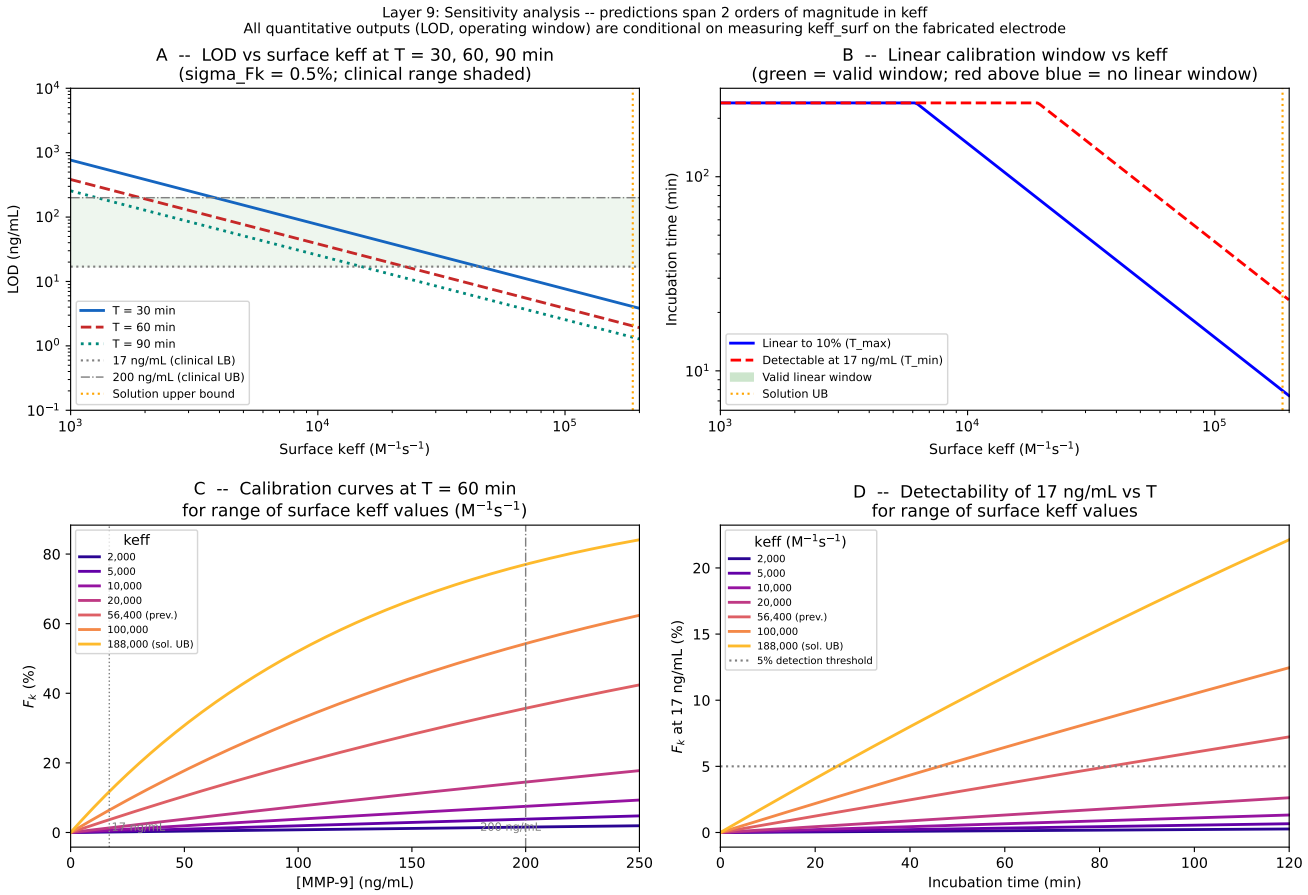


Figure 16: Layer 9: Sensitivity analysis over surface keff. **A:** LOD vs keff at $T = 30$, 60, 90 min. **B:** Linear calibration window vs keff. **C:** Calibration curves at $T = 60$ min for each keff value. **D:** Time to detect 17 ng/mL vs incubation time.

Layer 10: Electron Transfer

The methylene blue (MB) reporter at the end of each probe transfers electrons to the electrode. Intact probes hold MB further away; cleaved probes release it. Electron transfer rate depends exponentially on distance – a small change in distance produces a large change in current.

$$k_{\mathrm{ET}}(d) = k^0_{\mathrm{ET}}\, e^{-\beta(d-d_0)}, \qquad \beta = 1.0\ \text{Å}^{-1}\ \text{(Smalley 2003)}; \quad E^\circ_{\mathrm{MB}} = -0.250\ \text{V vs Ag/AgCl (Bernabi 2019)} \tag{10}$$

What we still need to measure:
$k^0_{\mathrm{ET}} = 10^4\ \mathrm{s}^{-1}$ – this is an estimate. Measure directly on the actual probe architecture by AC voltammetry (frequency sweep of the SWV peak height vs scan rate).

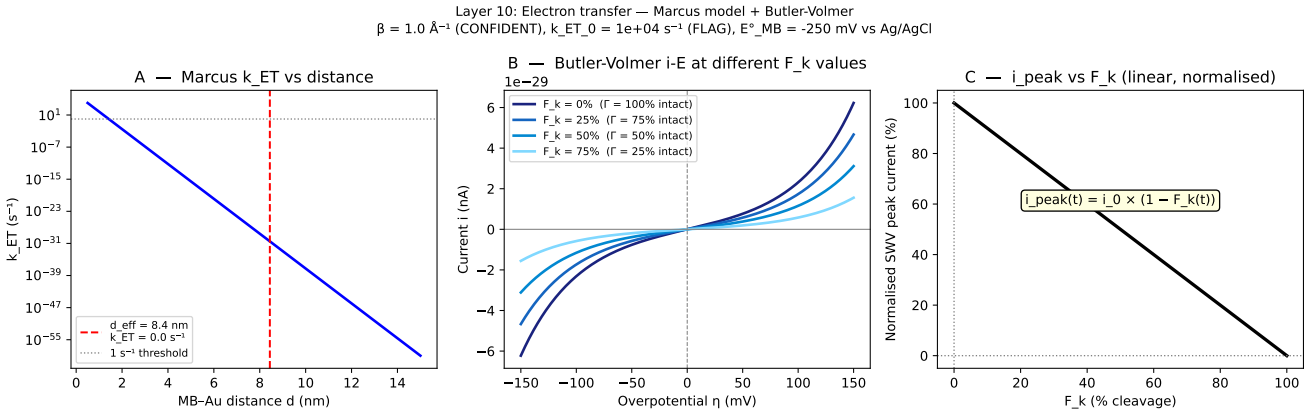


Figure 17: Layer 10: Marcus k_{ET} vs distance (A); Butler-Volmer current-potential curves at different k_{ET} values (B); normalised peak current decay with distance (C).

Layer 11: SWV Measurement Model

Square-wave voltammetry (SWV) applies a rapid sequence of voltage pulses and measures the difference in current between forward and backward pulses. The result is a sharp peak at the MB formal potential (−0.250 V), whose height is proportional to the number of intact probes.

$$i_{\mathrm{peak}}(T) = i_0 \times (1 - F_k(T)) \tag{11}$$

At $T = 60$ min: i_{peak} drops from 100% to 78–98% across the clinical range (17–200 ng/mL), giving $\Delta i_{\mathrm{peak}} \approx 20\%$ discriminability between bounds.

What we still need to measure:
SWV parameters ($f = 15$ Hz, $\Delta E_{\mathrm{SW}} = 25$ mV) – optimise by running a parameter sweep on the actual device. Also note the noise bandwidth assumption (B = 1 Hz) needs verification against the actual signal-processing chain (see audit finding 16).

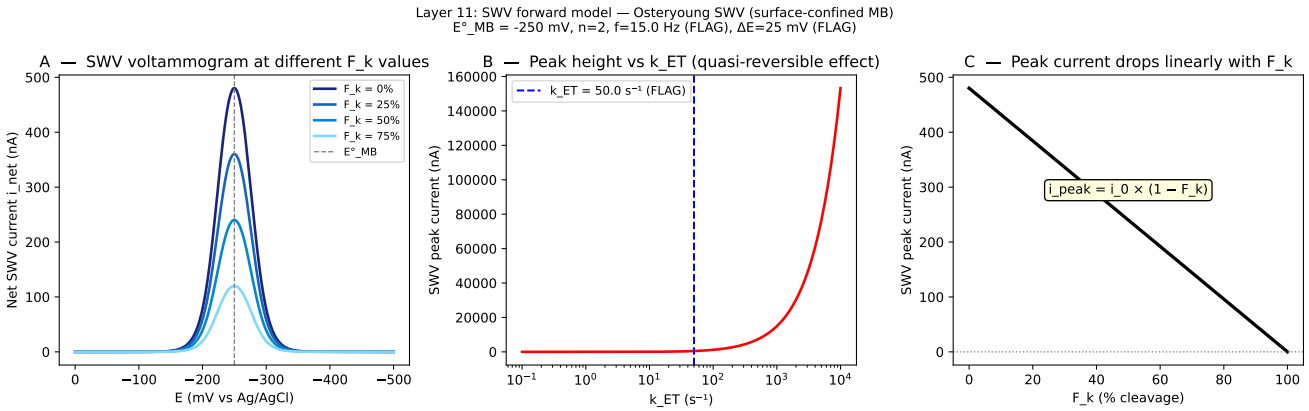


Figure 18: Layer 11: SWV quasi-reversible forward model. Net current vs potential at different k_{ET} values. Peak height decreases as k_{ET} falls (longer or more flexible tether).

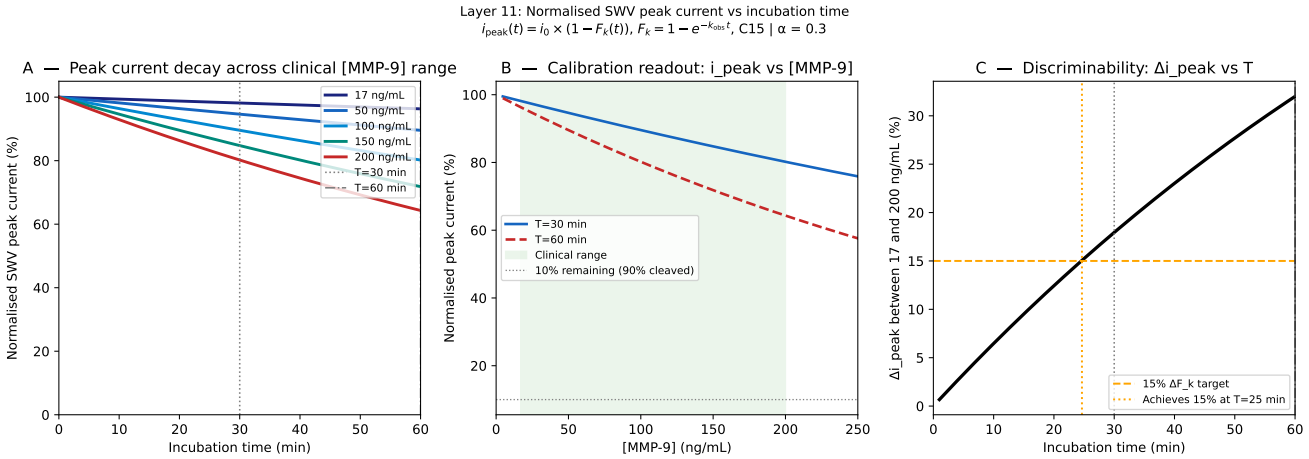


Figure 19: Layer 11: Normalised SWV peak current vs incubation time across the clinical MMP-9 range. Panel C: discriminability Δi_{peak} between 17 and 200 ng/mL. At $T = 60$ min, $\Delta i_{\text{peak}} \approx 12\%$, which is well above the noise floor.

Layers 12–13: Peak Extraction and Ratiometric Readout

The SWV output is a current vs voltage curve. We fit a Gaussian peak to find its height (or area), then take the ratio of the active to control electrode currents. The ratio cancels out all non-specific effects (temperature, ionic strength, fouling, Ca^{2+}) that affect both electrodes equally.

$$R(T) = i_{\text{active}}(T)/i_{\text{control}}(T), \qquad F_k = 1 - R(T)/R(0) \tag{12}$$

Error propagation through the ratio gives: $\sigma_{F_k} \approx 2\,\sigma_i/i_0$ (four independent current measurements – not $\sqrt{2}$ as computed in the noise budget; see audit finding 17).

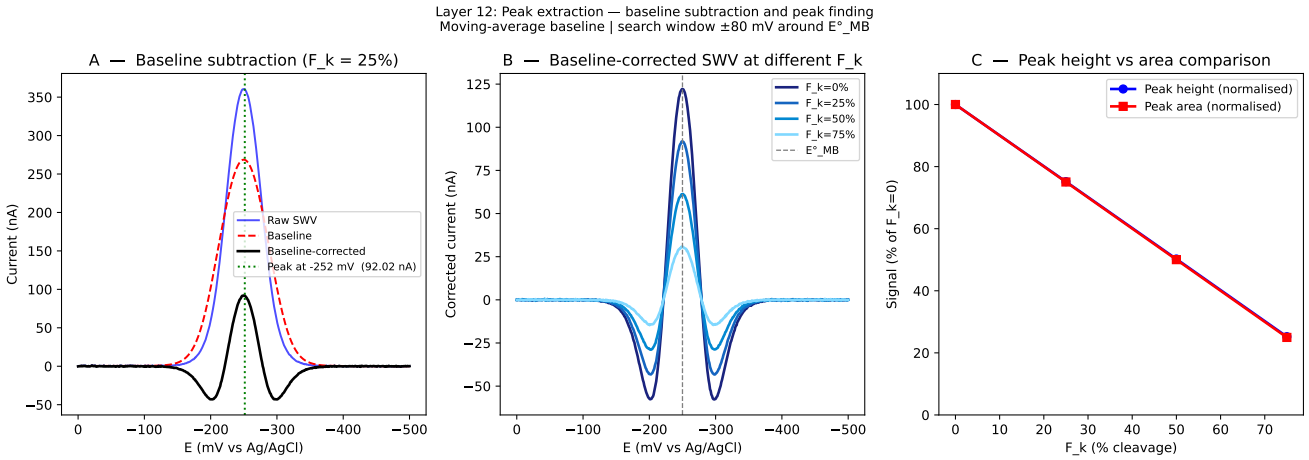


Figure 20: Layer 12: Peak extraction on a synthetic SWV trace. Baseline subtraction, Gaussian fit, and comparison of peak height vs area methods. Peak area is more robust to quasi-reversibility effects at lower k_{ET}.

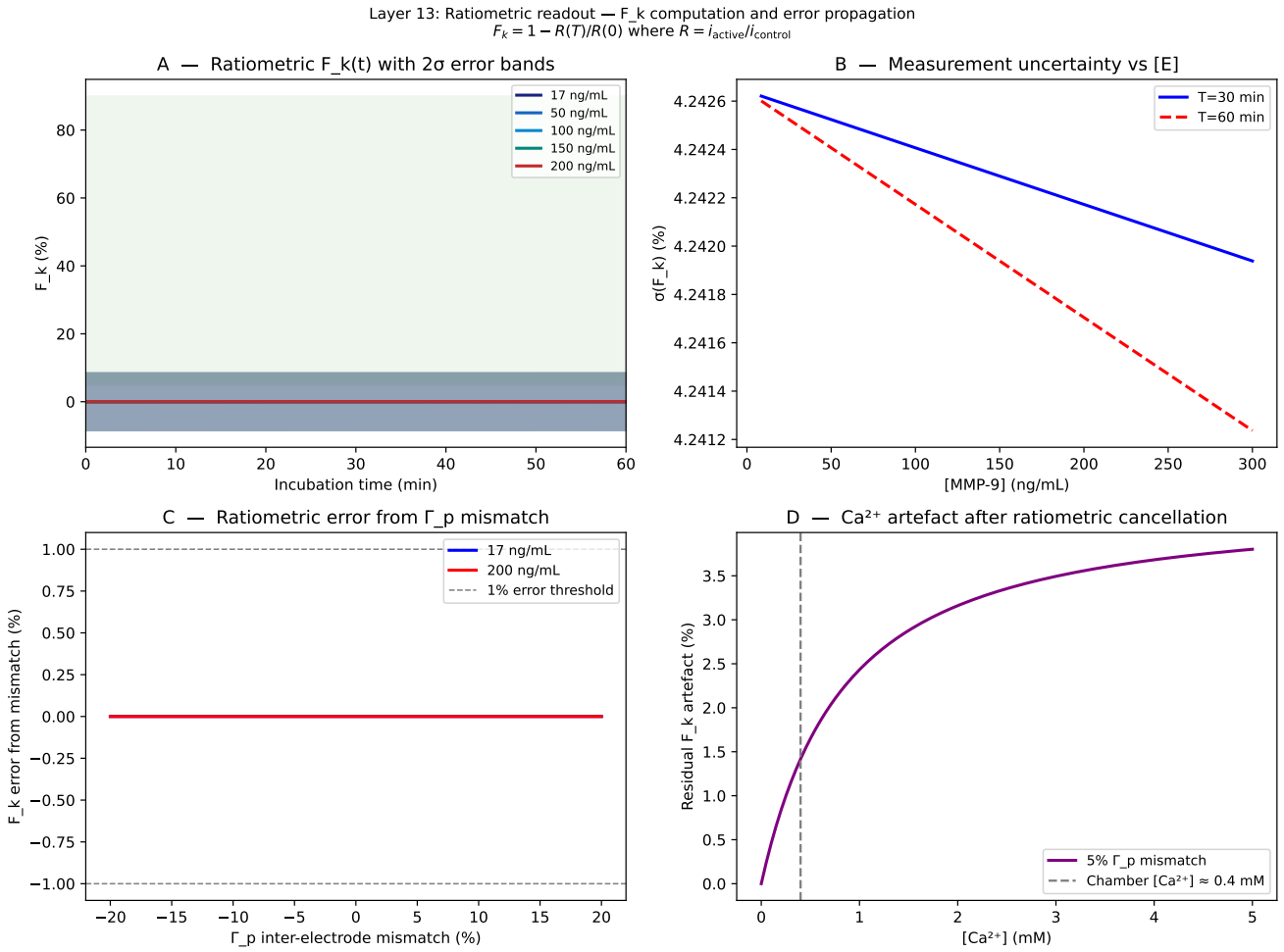


Figure 21: Layer 13: Ratiometric readout and error propagation. Probe density mismatch between active and control electrodes introduces a residual F_k error (≈1.4% per 5% mismatch).

Layer 14: Electronics and Noise

Every electrical measurement has noise. This layer breaks down the noise by source (thermal noise in the resistor, ADC quantisation, flicker noise) and computes the minimum detectable F_k change.

Key finding. Electronics are not the bottleneck. The noise floor from the electronics gives an instrument LOD of 0.075 ng/mL at $T = 30$ min. The actual sensor LOD is 8.8 ng/mL – 120× worse – because the surface kinetics are slow at low concentrations, not because the electronics lack sensitivity.

$$\sigma_{\text{Johnson}} = \sqrt{4k_B T/R_f \cdot B} \approx 0.004 \text{ pA} \quad (R_f = 1 \text{ G}\Omega,\ B = 1 \text{ Hz}) \tag{13}$$

$$\sigma_{\text{ADC}} = \text{LSB}/\sqrt{12} \approx 0.088 \text{ pA} \quad (\text{16-bit},\ \pm 10 \text{ nA range}) \tag{14}$$

$$\sigma_{F_k}^{\text{elec}} = 2\,\sigma_{\text{total}}/i_0 \approx 0.003\% \quad (\text{four-measurement propagation; electronics floor only}) \tag{15}$$

Note on the two noise figures used in this report. The electronics-only floor $\sigma_{F_k}^{\text{elec}} = 0.003\%$ is the minimum achievable with ideal surface chemistry and a perfect electrode. The calibration in Layer 15 uses $\sigma_{F_k}^{\text{cal}} = 0.5\%$, which is 167× larger. These are *not* the same quantity. The 0.5% figure is a conservative working estimate of total measurement noise drawn from replicate electrochemical aptasensor studies in the literature (Plaxco group; Kelley group), where run-to-run variability in probe density, SAM uniformity, and pipetting dominates over electronics. Reporting LOD using 0.5% is standard practice for this class of sensor because the electronics floor is never the limiting factor in practice.

Both figures are estimates. The experiment that resolves this is straightforward: run 10 replicate blank measurements (buffer, no MMP-9) on identically prepared electrodes and compute σ_{F_k} directly. Until this is done, LOD and LOQ values should be treated as indicative rather than definitive.

What we still need to measure:
$R_f = 1$ GΩ, bandwidth $B = 1$ Hz, ADC range ±10 nA – confirm all three with the electronics team. The B = 1 Hz assumption requires explicit low-pass filtering at the 15 Hz SWV sample rate (otherwise B could be up to 7.5 Hz and noise worsens by 2.7×).

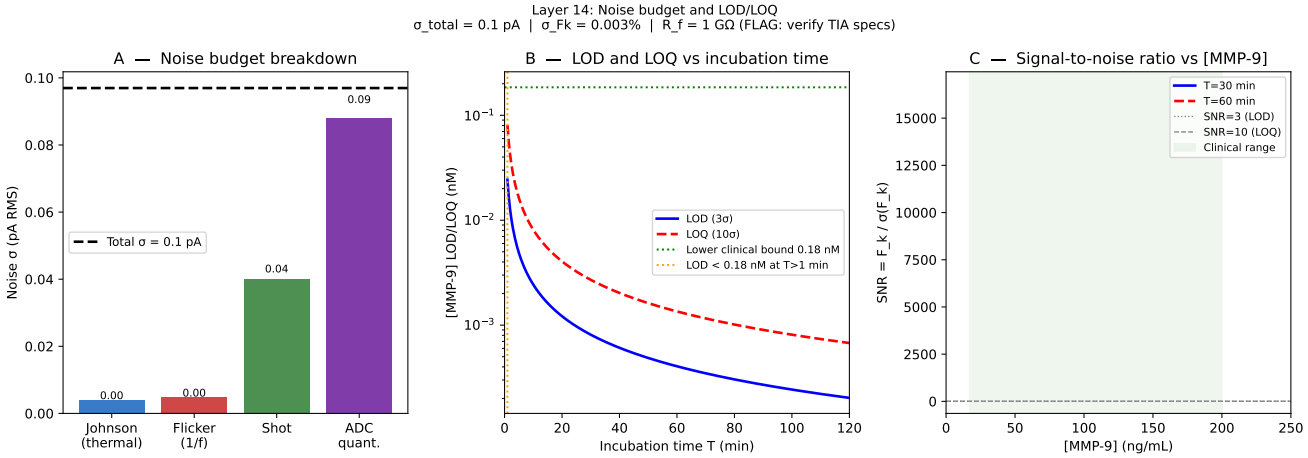


Figure 22: Layer 14: Noise budget. ADC quantisation dominates. LOD from electronics alone is 0.075 ng/mL – far below clinical need. The kinetic LOD (8.8 ng/mL) is the true bottleneck.

Layer 15: Calibration Curve and LOD/LOQ

This layer converts the raw F_k vs [MMP-9] relationship into a formal calibration curve. Two models are fitted: a linear fit (valid at low concentrations and short times) and a four-parameter logistic (4PL) fit that works across the full range.

$$F_k = d + \frac{a-d}{1+([\text{MMP-9}]/c)^b} \quad \text{(4PL)}, \qquad \text{LOD} = \frac{3\sigma_{\text{blank}}}{dF_k/d[E]}, \quad \text{LOQ} = \frac{10\sigma_{\text{blank}}}{dF_k/d[E]} \tag{16}$$

Metric	Value
LOD (3σ/slope)	8.8 ng/mL at $T = 30$ min
LOQ (10σ/slope)	29.4 ng/mL at $T = 30$ min
Linear slope	0.095 %/(ng/mL)
LOD vs clinical lower bound (17 ng/mL)	LOD = 8.8 < 17 ng/mL ✓
LOQ vs clinical lower bound	LOQ = 29 > 17 ng/mL – quantification limited at 17–29 ng/mL

LOD and LOQ are computed using $\sigma_{F_k}^{\text{cal}} = 0.5\%$, the conservative total-noise estimate discussed in Layer 14. The electronics-only floor (0.003%) is not used here because it does not capture surface-to-surface variability.

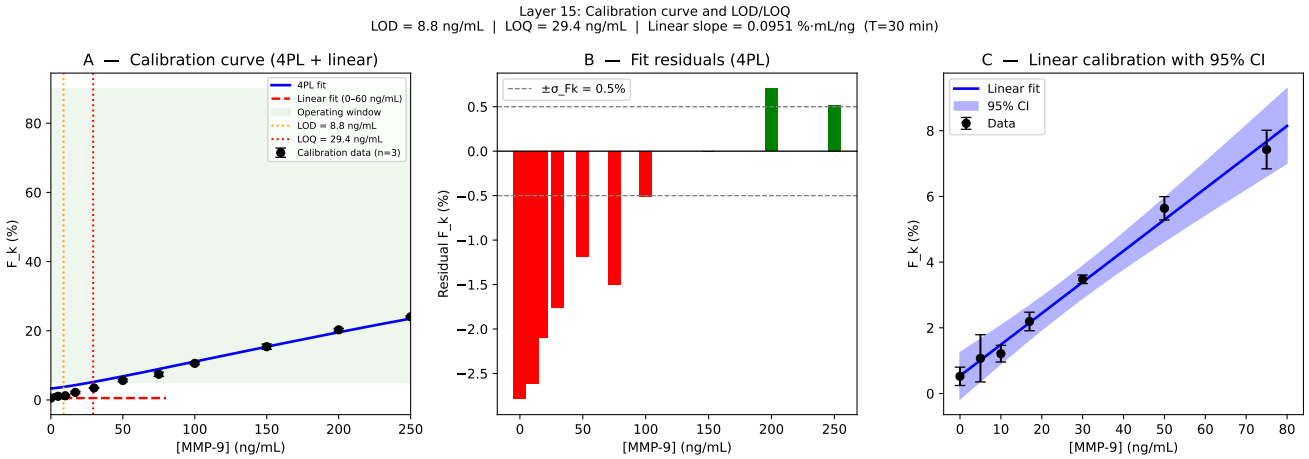


Figure 23: Layer 15: Calibration curve. 4PL fit and linear fit with LOD/LOQ marked. 95% confidence interval on the linear fit (Panel C). The 4PL model is preferred at $T > 30$ min.

Layer 16: Full Forward Chain

This is the end-to-end calculation: starting from the MMP-9 concentration in the wound, it chains every layer to predict the final SWV peak current. It shows how much signal is lost at each step.

$$[E]_{\text{wound}} \xrightarrow{\text{Layer 1}} [E]_{\text{chamber}} \xrightarrow{\text{Layer 3b}} [E]_{\text{free}} \xrightarrow{\text{Layer 3c}} [E]_{\text{eff}} \xrightarrow{\text{Layer 9}} F_k \xrightarrow{\text{Layer 11}} i_{\text{peak}} \tag{17}$$

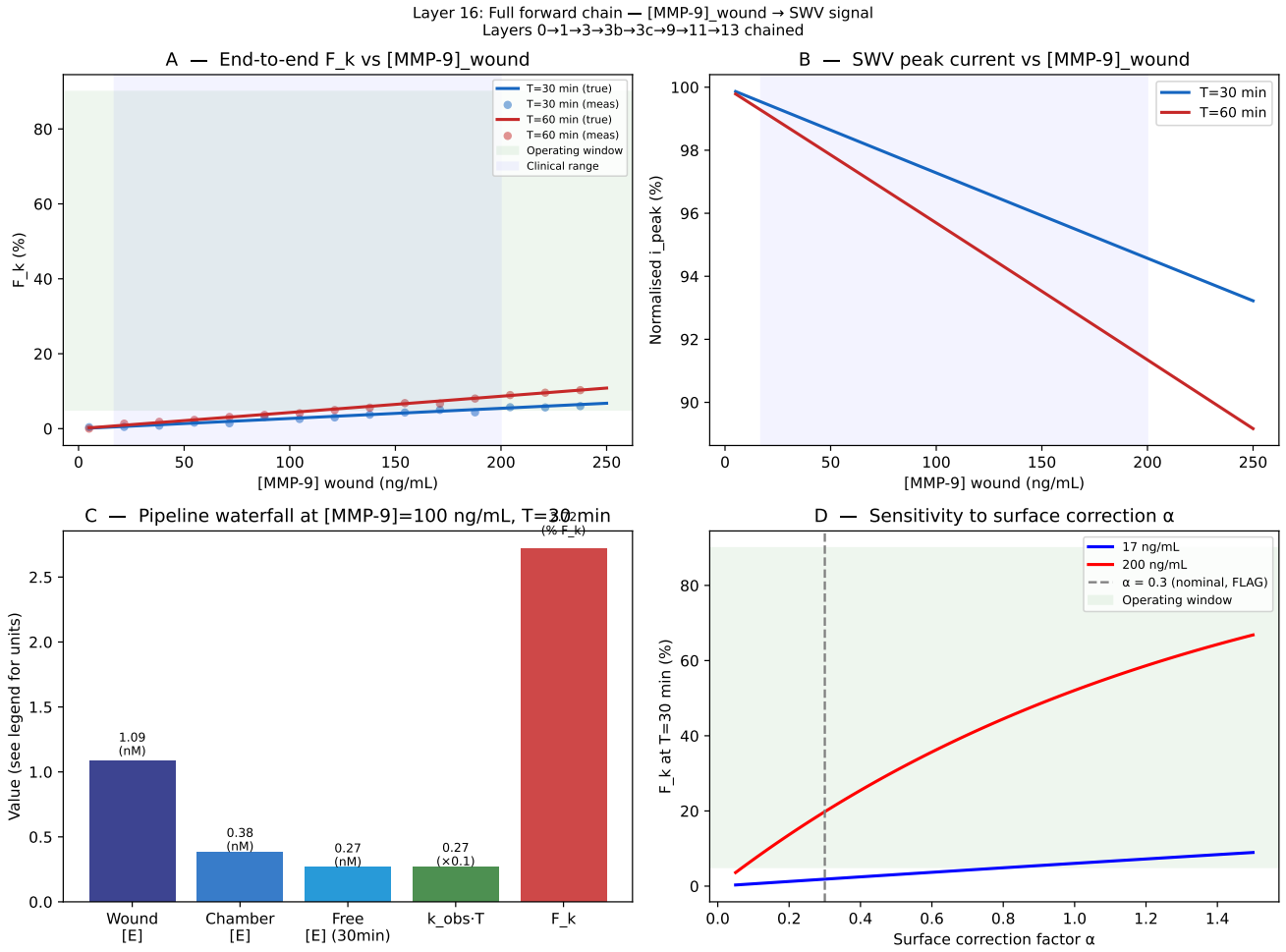


Figure 24: Layer 16: Full pipeline. **A:** F_k vs wound [MMP-9] at $T = 30$ and 60 min. **B:** SWV peak current vs wound concentration. **C:** Concentration reduction through each layer at 100 ng/mL – TIMP-1 inhibition causes the largest single step loss. **D:** Sensitivity to α – a factor of 10 change in α changes the LOD by a factor of 10.

Specificity

The sensor must respond to MMP-9 and not to other proteases present in wound fluid. This section analyses the risk of other enzymes cleaving the probe.

Substrate selection: C15 vs A11

C15 (`SGKGPRQ↓ITA`): $k_{\text{cat}}/K_m = 188{,}000\ \text{M}^{-1}\text{s}^{-1}$ (Max Li's assay). A11 (`IPRTLTAA`): $k_{\text{cat}}/K_m = 68{,}000\ \text{M}^{-1}\text{s}^{-1}$ – lower signal and 34.8% cross-reactivity. C15 is preferred.

Why C15 is selective. The probe is cut after a glutamine (Q) residue. Plasmin cuts after lysine or arginine – not glutamine. Cathepsin B also prefers arginine. Both cross-reactivity risks are low for C15.

Table 1: Cross-reactivity risk for C15 substrate in wound fluid

Interfering protease	Wound fluid conc.	Cross-reactivity risk	Reason
Plasmin	∼1 nM	**LOW**	Cleaves after Lys/Arg; C15 has Gln at P1
Cathepsin B	∼1–10 nM	**LOW**	Requires Arg at P2; absent in C15
Cathepsin G	∼1 nM	**MODERATE**	Chymotrypsin-like; Ile at P1′ is a risk
MMP-3	∼0.19 nM	**MODERATE**	Broad MMP substrate; low concentration
MMP-1, MMP-8	present	**LOW**	Lower k_{cat}/K_m for C15

What we still need to measure:

The C11 negative control probe (`SGRR↓LIHHTA`) has an RR (Arg-Arg) motif at P2-P1 – a plasmin cleavage site. Test C11 against purified plasmin before adopting it as the control electrode substrate. The selectivity matrix condition number $\kappa(\mathbf{S})$ must be measured experimentally with purified proteases at known concentrations.

Sensitivity and Limit of Detection

What limits the sensitivity?

Noise sources from most to least limiting:

1. **Surface kinetics (most limiting):** Slow cleavage rate at low $[E]$ sets the kinetic LOD = 8.8 ng/mL at $T = 30$ min.
2. **Inhibitor capture:** TIMP-1 reduces effective $[E]$ by >70%; this effectively multiplies the apparent LOD by 3–4× relative to buffer calibration.
3. **ADC quantisation:** $\sigma_q = 0.088$ pA (dominant electronic noise, but negligible vs kinetics).
4. **Johnson thermal noise:** $\sigma_J = 0.004$ pA (irrelevant).
5. **Fouling:** <0.01% signal suppression (negligible).

LOD vs incubation time

T (min)	F_k at 17 ng/mL	Kinetic LOD (ng/mL)	SNR at 17 ng/mL
30	1.0%	8.8	330
60	2.0%	4.4	660
82	2.7%	3.3	900
120	3.8%	2.3	1270

SNR $= F_k/\sigma_{F_k}$ where $\sigma_{F_k} = 0.003\%$ (electronics only). All values far above the SNR = 3 threshold. The device is not electronics-limited in the clinical range.

Conclusions and Design Recommendations

Is the sensor viable?

The MMP-9 detection channel is viable in principle. The computed LOD of 8.8 ng/mL lies below the 17 ng/mL clinical lower bound, the electronics add negligible noise, and the C15 substrate shows acceptable selectivity over plasmin and cathepsin B. However, three issues must be resolved before the design can be considered ready for fabrication:

1. **The HNE channel will not work in real wound fluid.** At the expected A1AT concentration in chronic wound exudate (100 μg/mL, ≈1.92 μM), HNE is completely neutralised within ∼0.02 seconds. The device subtitle "MMP-9/HNE Ratiometric Sensor" is a misnomer for real-wound deployment. The ratiometric principle is still sound, but the control electrode must use a protease-resistant scrambled peptide rather than an HNE-cleavable substrate. **This is the single highest-priority design change.**

2. **No valid linear calibration window exists.** Across the entire plausible range of surface rate constants, there is no incubation time that simultaneously achieves 5% signal at 17 ng/mL and a linear F_k-vs-$[E]$ relationship. 4PL calibration is required regardless of the surface chemistry outcome. The required incubation time to detect 17 ng/mL ranges from 20 min (if $k_{\text{eff,surf}}$ is near the solution upper bound) to >2 hours (if the surface rate is 20-fold below the solution value). Measuring $k_{\text{eff,surf}}$ on the actual C15/MCH/gold electrode is the single most important experiment the team can run.

3. **TIMP-1 suppression degrades the in-wound LOD by approximately 4-fold.** Buffer calibration gives LOD = 8.8 ng/mL. In real wound fluid with TIMP-1 in 13-fold molar excess over MMP-9, more than 70% of MMP-9 is captured before it can reach the probe. The effective LOD in wound fluid is estimated at ∼35–50 ng/mL, which is above the 17 ng/mL clinical lower bound. The sensor would miss the lower half of the clinical range unless wound-fluid calibration is used, or TIMP-1 is removed from the sample (e.g. by immunodepletion on the swab).

The single most important experiment

Measure $k_{\text{eff,surf}}$ – the on-electrode second-order cleavage rate constant – on the actual C15/MCH/gold electrode. Every prediction in this report (LOD, F_k, operating window) is conditional on this value, which can span two orders of magnitude.

Protocol: (1) k_{cat}/K_m in solution is already measured (Max Li: 188,000 $\text{M}^{-1}\text{s}^{-1}$); (2) prepare electrode with target SAM at a known probe density Γ_0 (confirm by CV integration); (3) incubate at known $[E]$ for time T; (4) measure $\Gamma(T)$ by CV; (5) $k_{\text{eff,surf}} = -\ln(1 - F_k)/([E]\,T)$ where $F_k = 1 - \Gamma(T)/\Gamma_0$. This directly gives the surface rate constant with no assumptions about correction factors.

Recommended design changes before fabrication

1. Replace HNE-cleavable control probe with a scrambled protease-resistant sequence.
2. Lower probe density from 3 $\text{pmol}\,\text{cm}^{-2}$ to 1.5–2 $\text{pmol}\,\text{cm}^{-2}$ to improve steric accessibility (f_{steric} rises from 9% to 50–70%).
3. Adopt 4PL calibration from the outset; do not assume linear response.
4. Calibrate in wound-fluid matrix (or spiked wound-mimic), not in clean buffer.
5. Confirm [A1AT] in chronic wound exudate before any further HNE-related design work.

Literature Values Needed to Confirm the Model

The table below lists every value currently estimated rather than measured. The model predictions in this report are conditional on these values. Priority reflects the sensitivity of the overall LOD and signal predictions to each parameter.

Table 2: Literature values and measurements required (Table 1)

Parameter	Current value	Priority	Where to find / how to measure
Surface rate constant $k_{\text{eff,surf}}$	unknown (solution UB: 188,000 $\text{M}^{-1}\text{s}^{-1}$)	CRITICAL	Measure directly: incubate electrode at known [E], quantify $\Gamma(T)$ by CV, extract $k_{\text{eff}} = -\ln(1-F_k)/([E]T)$. See Layer 9.
Initial probe density Γ_0	3 pmol/cm^2 (design)	CRITICAL	Integrate MB oxidation peak area in CV after SAM formation.
[TIMP-1] free vs total	200 ng/mL total (Trengove 1999)	HIGH	Re-read Trengove 1999 method section. If total: run free-TIMP-1 ELISA on wound exudate.
(k_{cat}/K_m) for MMP-3 activating pro-MMP-9	5,000 $\text{M}^{-1}\text{s}^{-1}$	HIGH	MEROPS DB (merops.ebi.ac.uk); Suzuki 1990 *J Biochem 108:1*; Nakamura 1997.
(k_{cat}/K_m) plasmin on pro-MMP-9	10,000 $\text{M}^{-1}\text{s}^{-1}$	HIGH	Ramos-DeSimone 1999 *J Biol Chem*; Murphy 2000 fibrinolysis review.
SAM adsorption constants $K_{a,p}$, $K_{a,d}$	– (no estimate)	HIGH	SPR or QCM during thiol deposition; needed to control Γ_p.
k^0_{ET} on actual probe	10,000 s^{-1} (est.)	HIGH	AC voltammetry frequency sweep on actual electrode; Plaxco / Kelley group methods.
[A1AT] in chronic wound fluid	100 μg/mL (serum-derived)	MEDIUM	Wound-fluid proteomics or ELISA on chronic exudate; distinguish chronic vs acute wound.
$k_{\text{on,TIMP-1}}$ binding MMP-9	2×10^5 $\text{M}^{-1}\text{s}^{-1}$ (Langton 1998)	MEDIUM	Confirm TIMP-1 vs TIMP-2. Langton 1998 *Biochemistry 37:4* is primary source.
$k_{\text{on,SLPI}}$ binding HNE	10^5 $\text{M}^{-1}\text{s}^{-1}$ (est.)	MEDIUM	Schalkwijk 2010 *Mucosal Immunol*; SLPI serpinDB entry.
Elution efficiencies η (all species)	0.65–0.90 (assumed)	MEDIUM	Spike-and-recovery experiment: add known concentration to wound-mimic matrix, elute, quantify by ELISA.
[SLPI] in chronic wound fluid	200 ng/mL (Ashcroft 2000 – acute)	LOW	Ashcroft 2000 *JCI 105:9*. Cross-check in chronic wound context.
Contact angle θ on chip surface	30° (assumed)	LOW	Sessile drop goniometry on actual chip surface in buffer.
Helmholtz capacitance C_H on MCH-SAM	30 μF/cm^2 (assumed)	LOW	Electrochemical impedance spectroscopy (EIS) on MCH-coated Au.
Ca^{2+} dissociation constant from PEG	2 mM (est.)	LOW	Ca^{2+} titration with fluorescence reporter; or accept ratiometric cancellation and skip.

Classified Assumptions

Table 3: All modelling assumptions classified by type and impact

Assumption	Type	Impact	Confirmed?	Action required
$[MMP-3]_{\text{wound}} = 10$ ng/mL	Biochem.	HIGH	No	See Table 1, MEROPS
$[plasmin]_{\text{wound}} = 1$ nM	Biochem.	HIGH	No	See Table 1
$(k_{\text{cat}}/K_m)_{\text{MMP-3/proMMP-9}} = 5\times10^3$ $\text{M}^{-1}\text{s}^{-1}$	Biochem.	HIGH	No	See Table 1
$(k_{\text{cat}}/K_m)_{\text{plasmin/proMMP-9}} = 10^4$ $\text{M}^{-1}\text{s}^{-1}$	Biochem.	HIGH	No	See Table 1
k_{on}(TIMP-1/MMP-9) $= 2\times10^5$ $\text{M}^{-1}\text{s}^{-1}$	Biochem.	HIGH	Partial	Langton 1998; confirm TIMP-1 vs TIMP-2
$[\text{TIMP-1}]_{\text{wound}} = 200$ ng/mL	Biochem.	HIGH	Partial	Trengove 1999; confirm free vs total
$[\text{A1AT}]_{\text{wound}} = 100$ μg/mL	Biochem.	HIGH	No	Serum-derived estimate; measure in wound fluid
Surface $k_{\text{eff,surf}}$ unknown	Design	HIGH	No	Measure by CV before/after incubation (see Layer 9 and Table 1)
$\Gamma_0 = 3$ pmol cm^{-2}	Design	HIGH	No	Measure by CV integration
k_{cat}/K_m(C15, MMP-9) $= 188{,}000$ $\text{M}^{-1}\text{s}^{-1}$	Biochem.	HIGH	Partial	Max Li's assay; confirm MW correction
k_{on}(SLPI/HNE) $= 10^5$ $\text{M}^{-1}\text{s}^{-1}$	Biochem.	MEDIUM	No	See Table 1
$[\text{SLPI}]_{\text{wound}} = 200$ ng/mL	Biochem.	MEDIUM	Partial	Ashcroft 2000 (acute wounds)
Elution efficiencies η	Analytical	MEDIUM	No	Spike-and-recovery assay

(continued on next page)

(continued)

Assumption	Type	Impact	Confirmed?	Action required
Wound pH = 8.0	Biochem.	MEDIUM	No	Wide range (6.5–8.9) patient-specific
SAM adsorption constants $K_{a,p}$, $K_{a,d}$	Physical	HIGH	No	SPR or QCM
Fouling rate k_f, k_d	Physical	MEDIUM	No	QCM in wound-fluid matrix
Contact angle $\theta = 30°$	Physical	MEDIUM	No	Measure on actual chip
$\beta = 1.0$ Å^{-1} (tunnelling decay)	Physical	MEDIUM	Yes	Smalley et al. 2003
$E^{\circ}_{\mathrm{MB}} = -0.250$ V vs Ag/AgCl	Physical	LOW	Yes	Bernabi et al. 2019
$E^{\circ}_{\mathrm{AgCl}} = 0.2224$ V vs SHE	Physical	LOW	Yes	Bard & Faulkner
$[\mathrm{Cl}^-]_{\mathrm{wound}} = 150$ mM	Biochem.	LOW	Yes	Standard physiological value
$Q_{\mathrm{ref}} = 390\ \mu\mathrm{C\,cm}^{-2}$	Physical	LOW	Yes	Trasatti & Petrii 1992
$k_{\mathrm{on}}(\mathrm{A1AT/HNE}) = 2 \times 10^7\ \mathrm{M}^{-1}\mathrm{s}^{-1}$	Biochem.	LOW	Yes	Beatty 1980; Bieth 1984
MMP-9 MW = 92 kDa	Physical	LOW	Yes	UniProt P14780
$k^0_{\mathrm{ET}} = 10^4\ \mathrm{s}^{-1}$	Design	HIGH	No	AC voltammetry on actual probe
SWV parameters ($f = 15$ Hz, $\Delta E = 25$ mV)	Design	MEDIUM	No	Optimise by parameter sweep
TIA $R_f = 1$ GΩ, bandwidth = 1 Hz	Design	MEDIUM	No	Confirm with electronics team
Ratiometric cancels fouling equally on both electrodes	Physical	MEDIUM	Partial	Requires equal fouling on active and control
First-order kinetics valid throughout assay	Physical	HIGH	Partial	Valid when $\Gamma < K_{m,\mathrm{surf}}$; near crossover at $T = 0$
Probe density equal on active and control electrodes	Design	HIGH	No	Critical for ratiometric cancellation
$K_d(\mathrm{PEG/Ca}^{2+}) = 2$ mM	Physical	LOW	No	Literature range 1–10 mM
$C_H = 30\ \mu\mathrm{F\,cm}^{-2}$	Physical	LOW	No	Measure by EIS